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Batey et al.

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(54) **COMPOSITIONS AND METHODS OF USE FOR SMALL-MOLECULE REGULATION OF CRISPR-CAS9 ACTIVITY USING RNA APTAMERS**

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(71) Applicants: **THE REGENTS OF THE UNIVERSITY OF COLORADO, A BODY CORPORATE**, Denver, CO (US); **INSCRIPTA INC.**, Boulder, CO (US)

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Primary Examiner — Evelyn Y Pyla
Assistant Examiner — Katherine R Small
 (74) *Attorney, Agent, or Firm* — Polsinelli PC

(72) Inventors: **Robert Batey**, Boulder, CO (US); **Roman S. Iwasaki Cordero**, Boulder, CO (US); **Andrew Garst**, Boulder, CO (US)

(73) Assignees: **The Regents of the University of Colorado, a body corporate**, Denver, CO (US); **Inscripta Inc.**, Boulder, CO (US)

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(60) Provisional application No. 62/793,748, filed on Jan. 17, 2019.

(51) **Int. Cl.**
C12N 15/115 (2010.01)
C12N 9/22 (2006.01)
C12N 15/11 (2006.01)

(52) **U.S. Cl.**
 CPC **C12N 15/115** (2013.01); **C12N 9/22** (2013.01); **C12N 15/111** (2013.01); **C12N 2310/16** (2013.01); **C12N 2310/20** (2017.05)

(58) **Field of Classification Search**
 CPC C12N 15/115; C12N 9/22; C12N 15/111
 See application file for complete search history.

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(57) ABSTRACT

Provided herein are single guide RNAs (sgRNAs) that comprise aptamer sequences and related compositions and methods. Also provided herein are methods of selecting inducible sgRNAs that comprise aptamer sequences.

19 Claims, 14 Drawing Sheets

Specification includes a Sequence Listing.

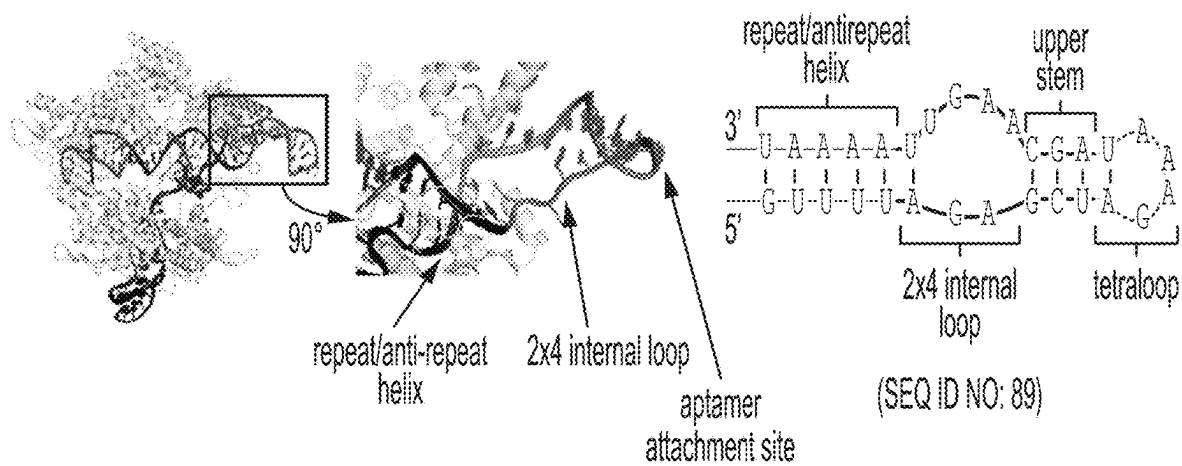


FIG. 1A

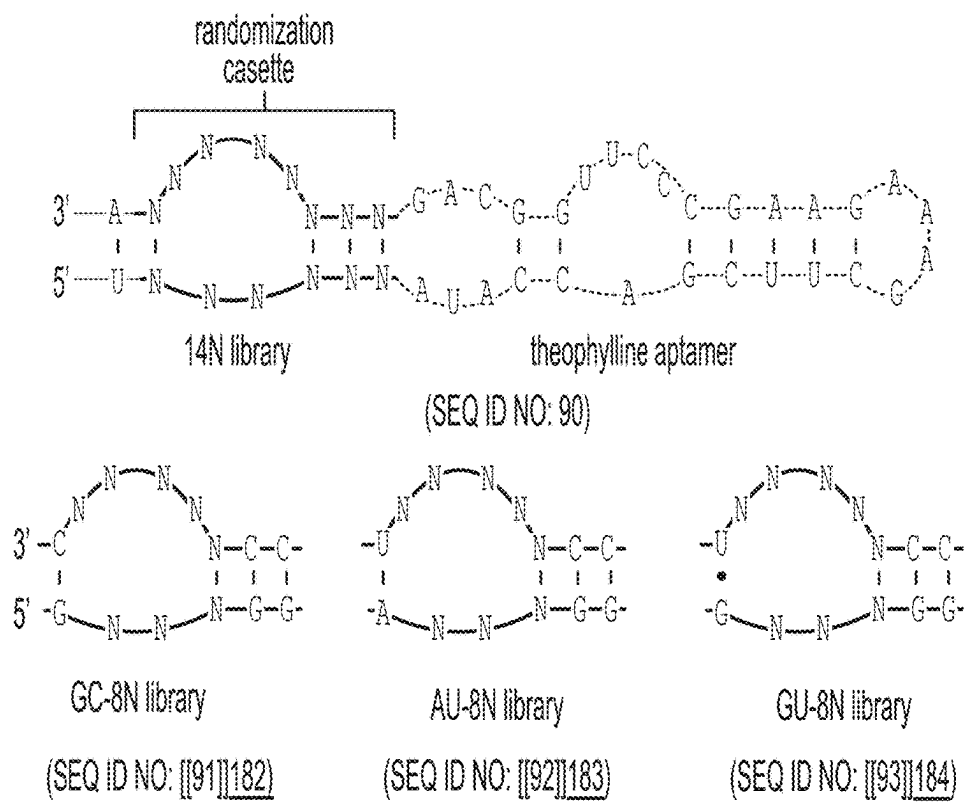


FIG. 1B

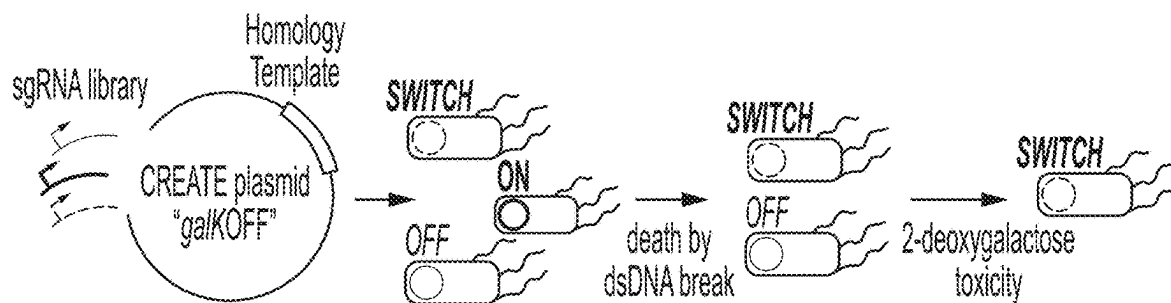


FIG. 1C

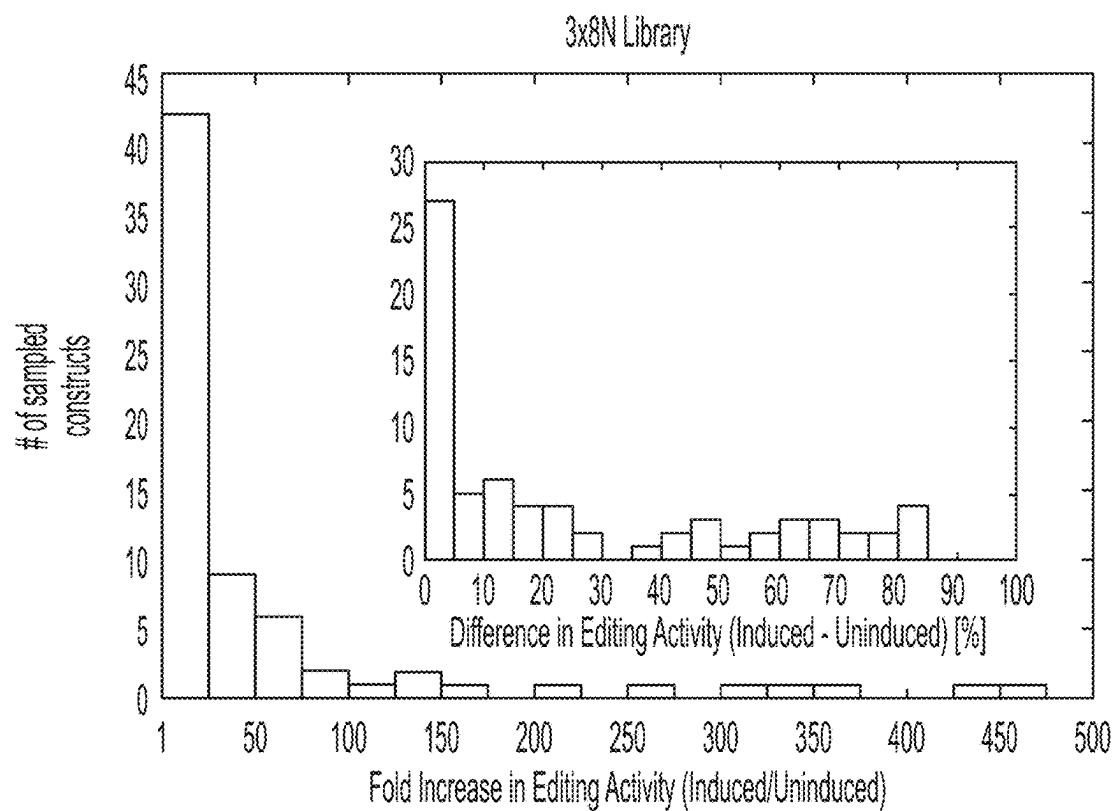


FIG. 1D

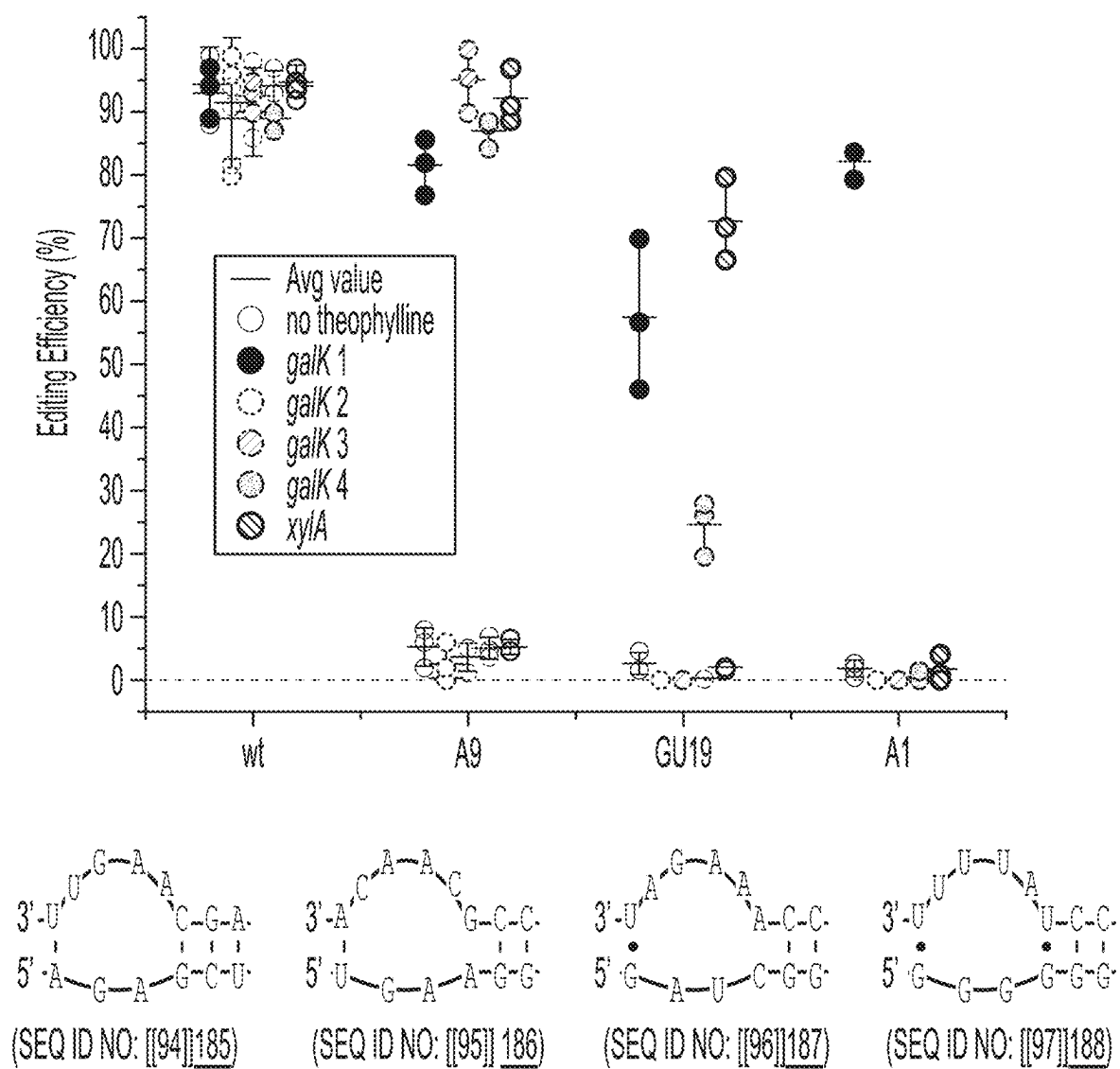


FIG. 1E

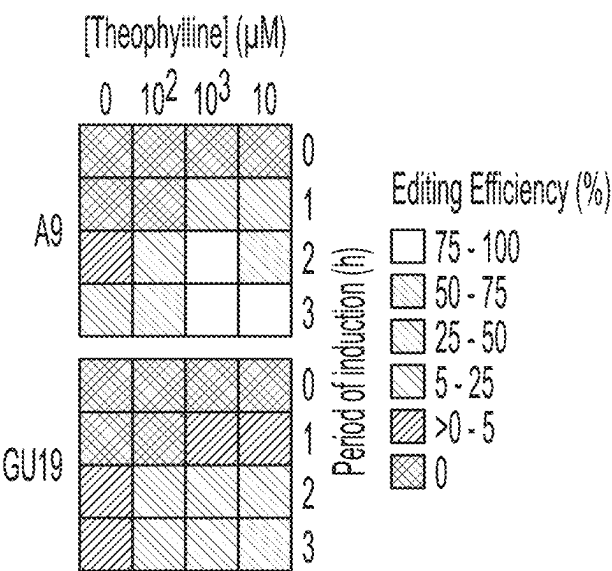


FIG. 2A

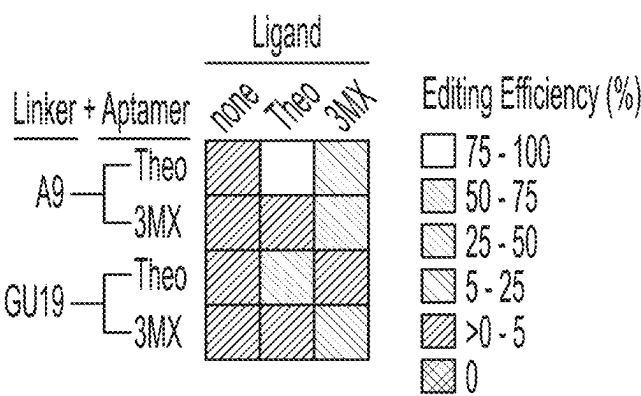


FIG. 2B

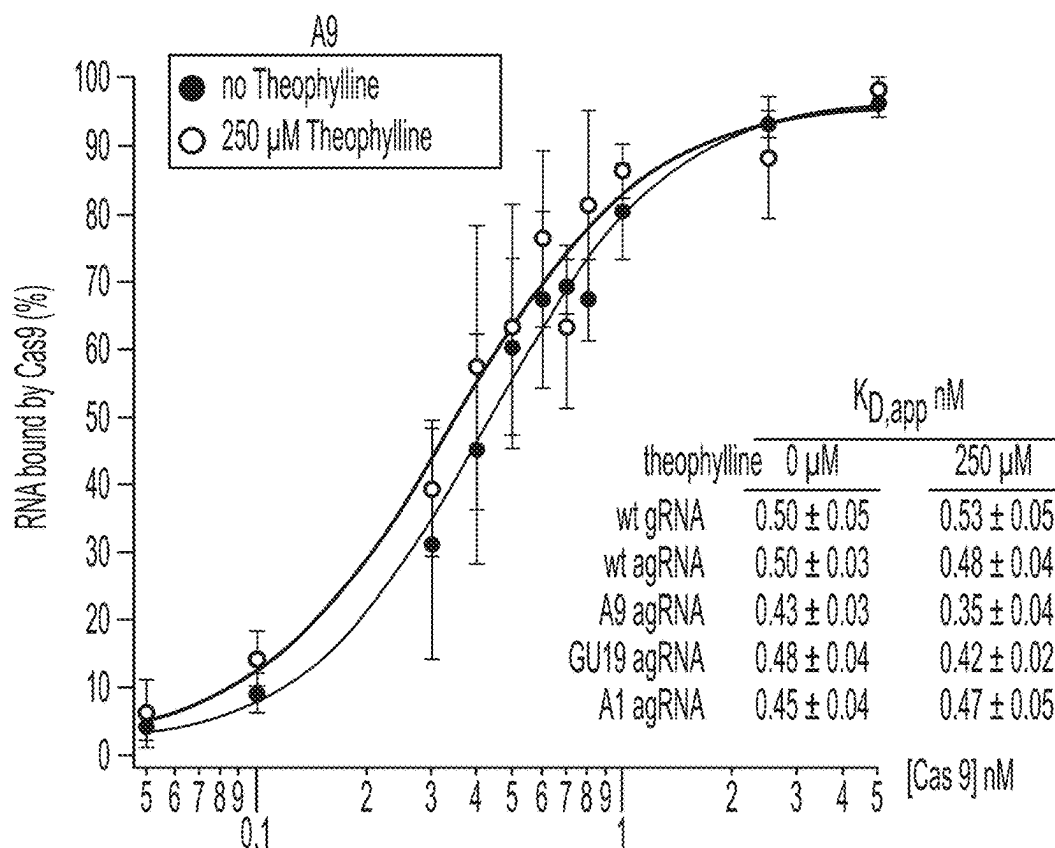


FIG. 2C

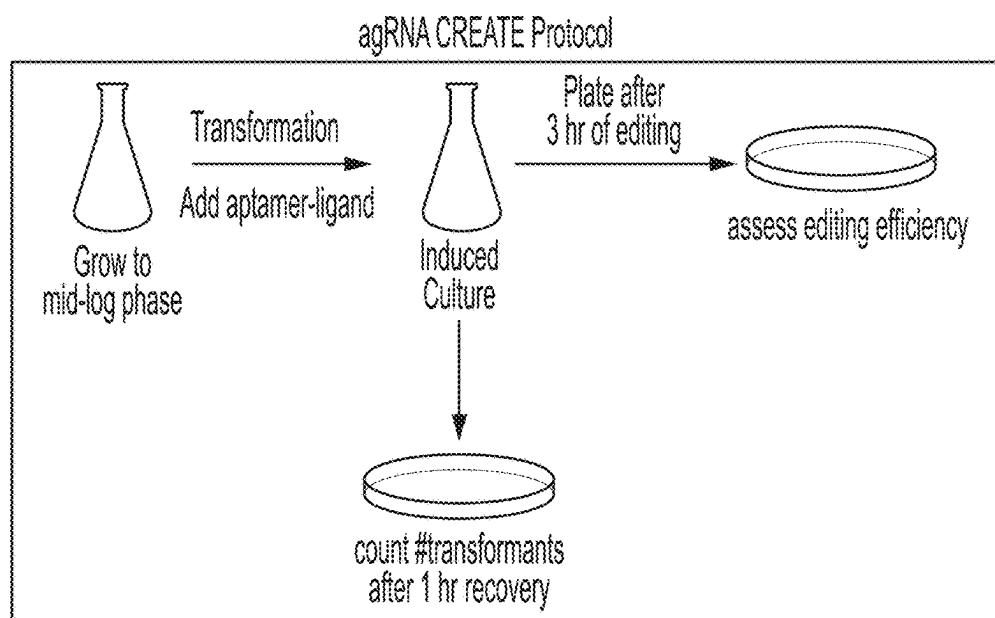


FIG. 3A

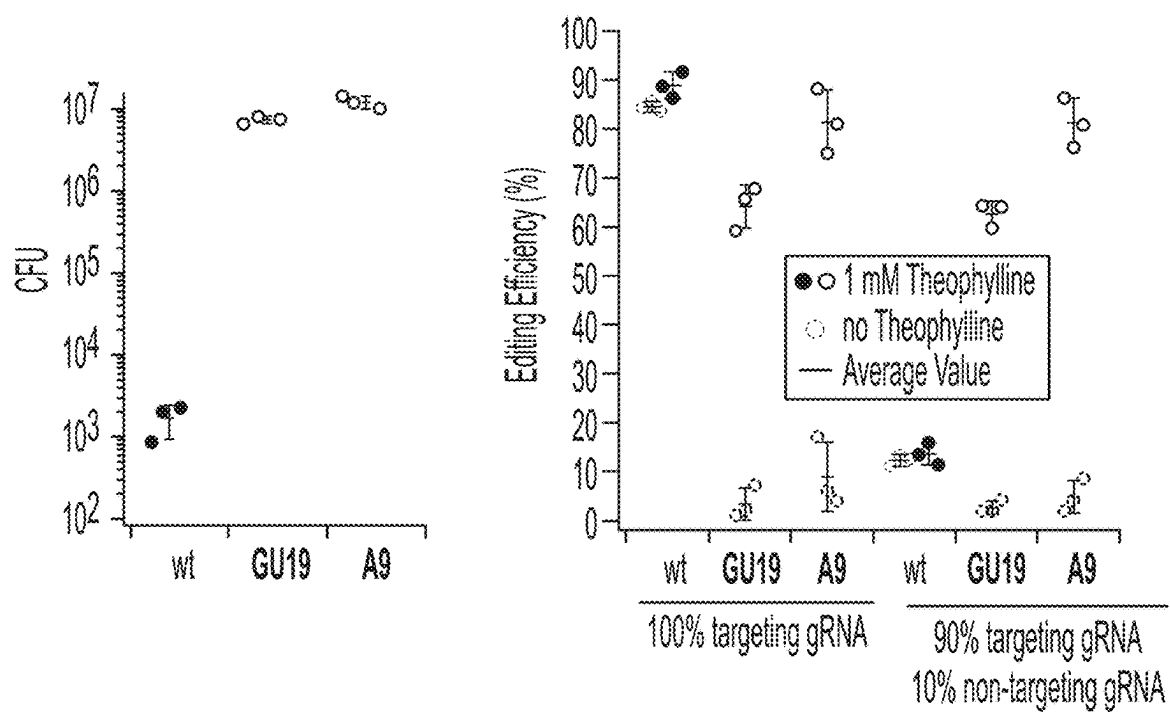


FIG. 3B

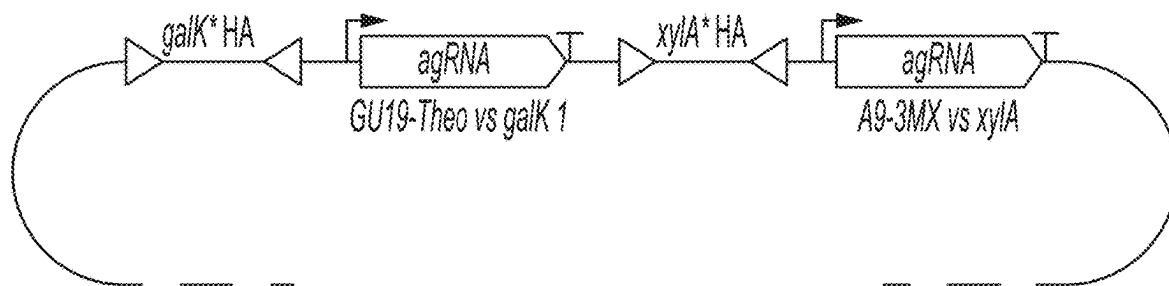


FIG. 3C

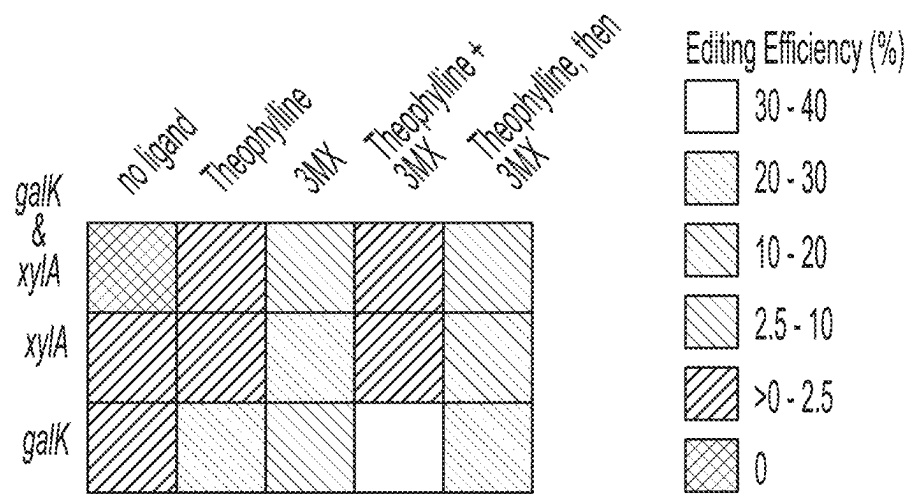


FIG. 3D

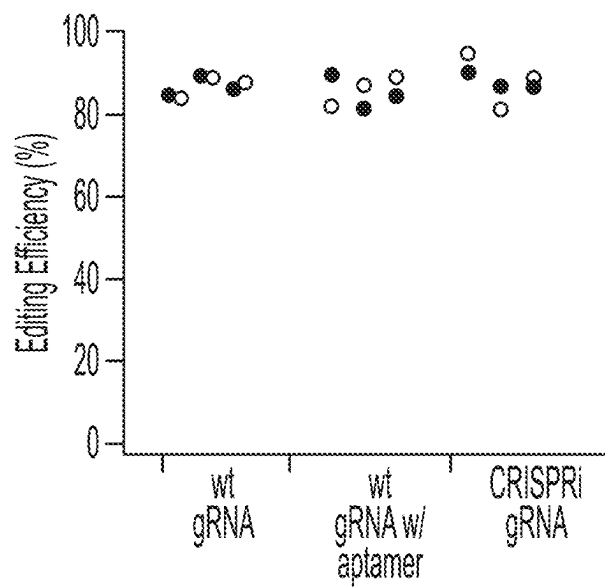


FIG. 4A

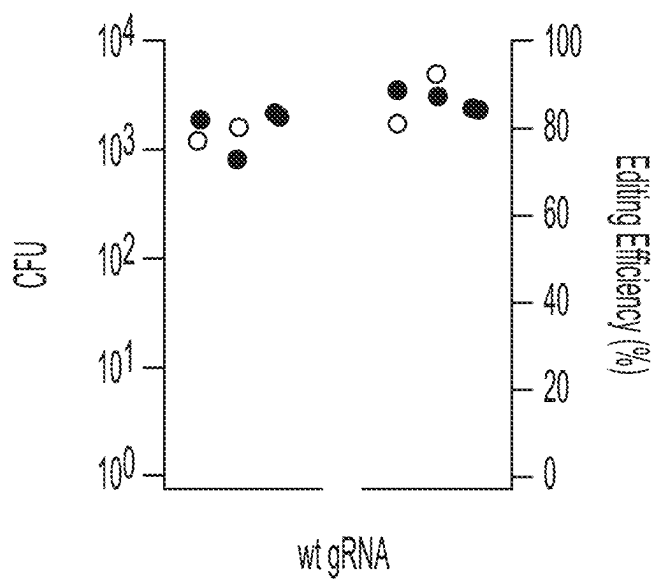


FIG. 4B

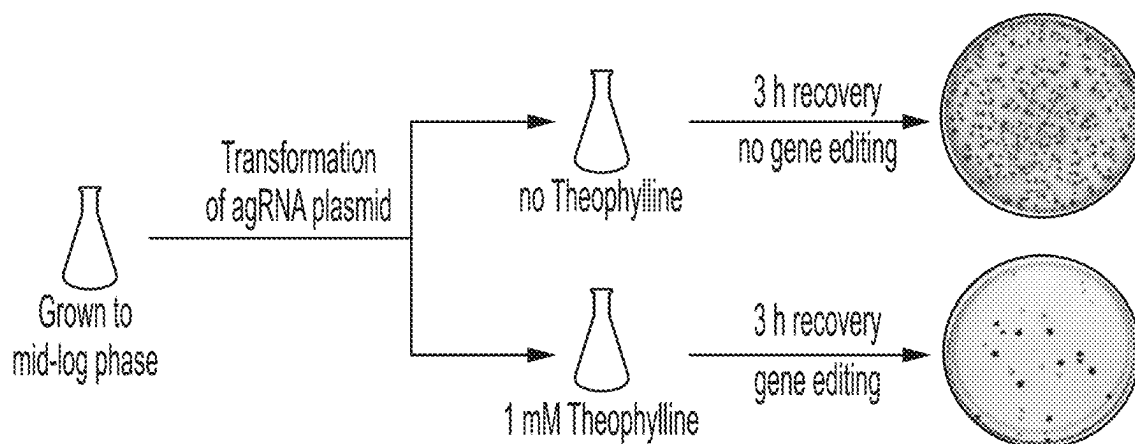


FIG. 5

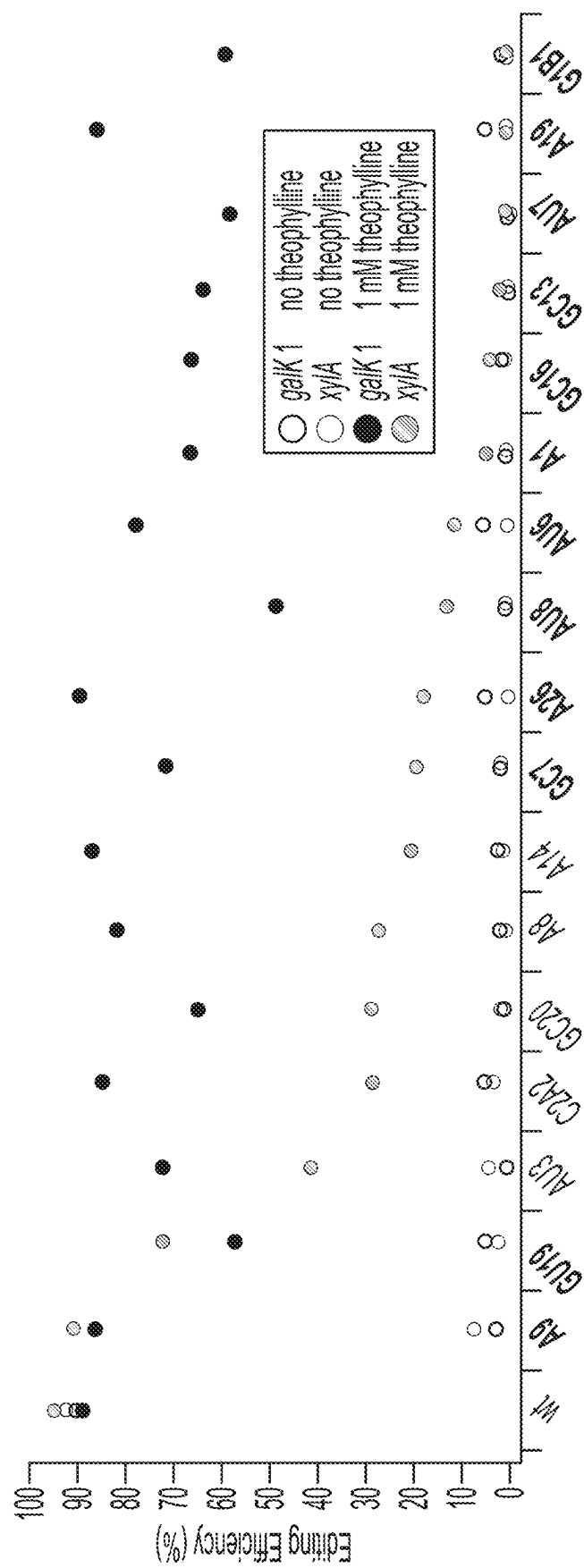


FIG. 6

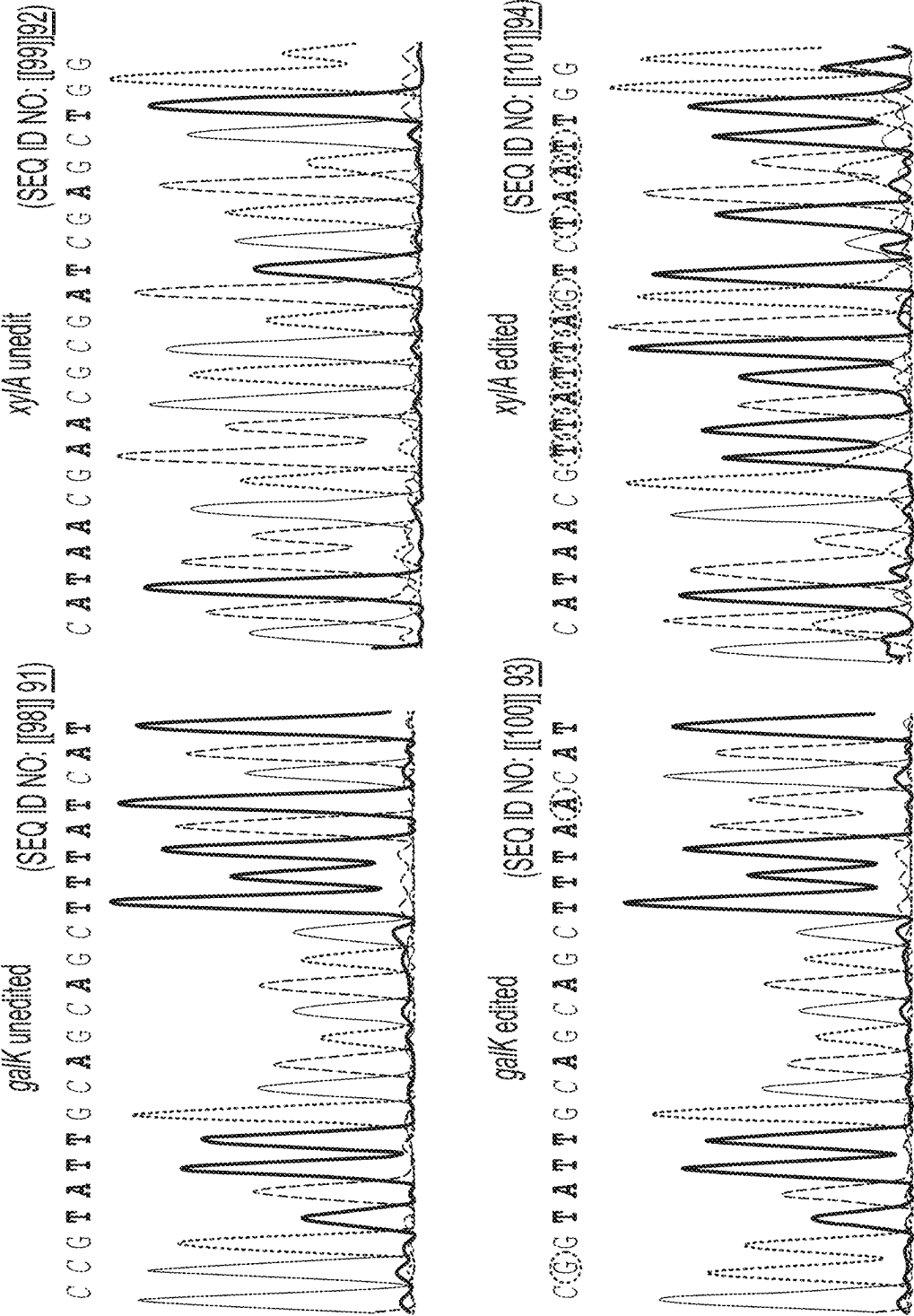


FIG. 7

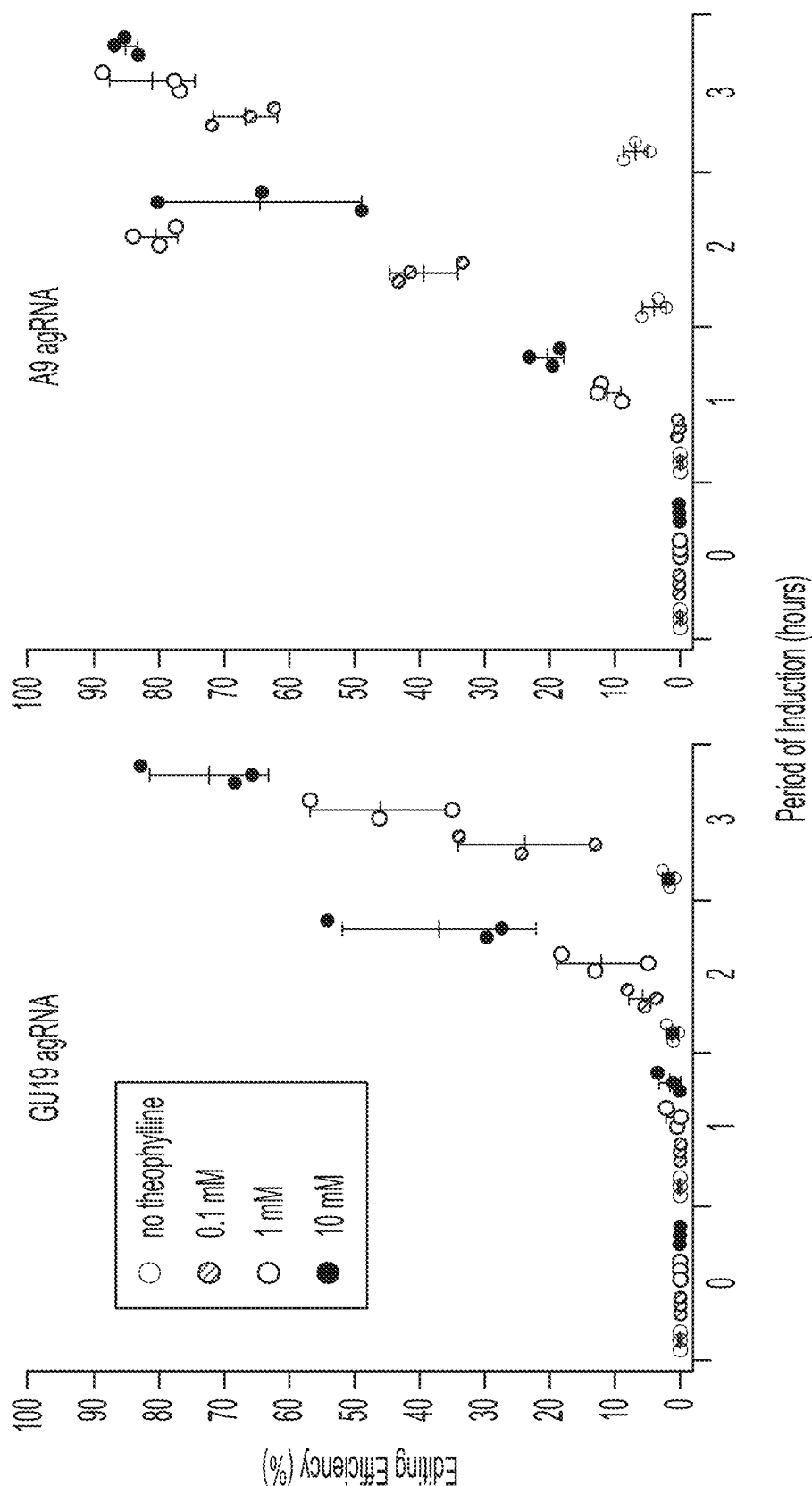


FIG. 8

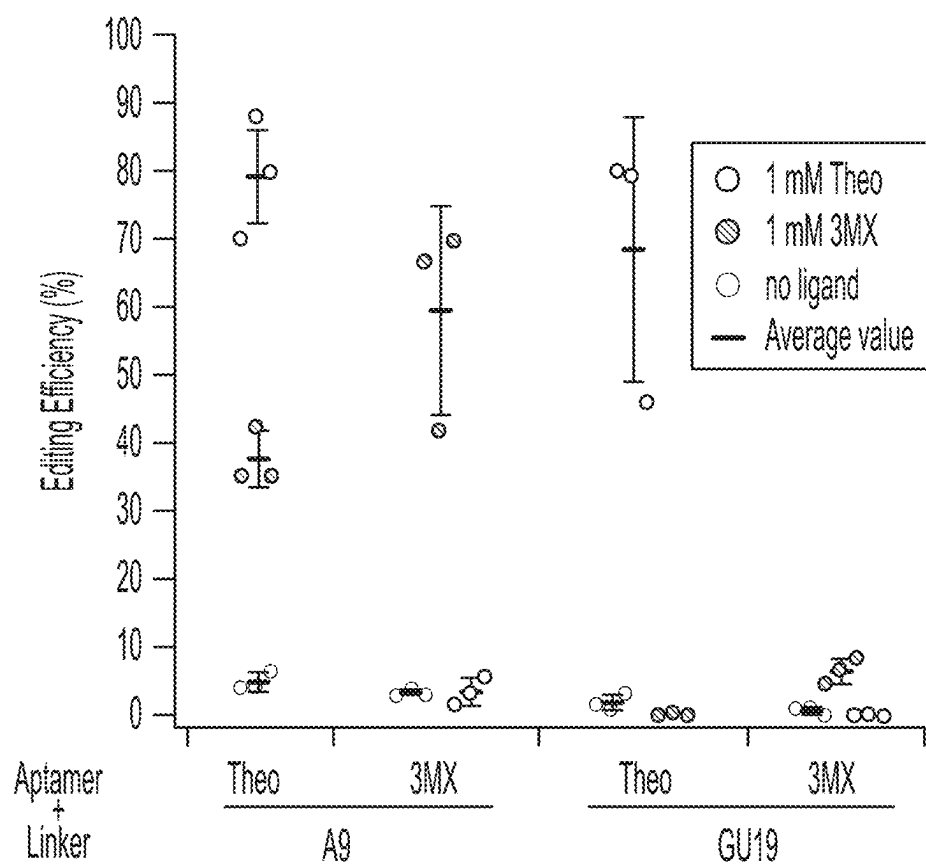


FIG. 9

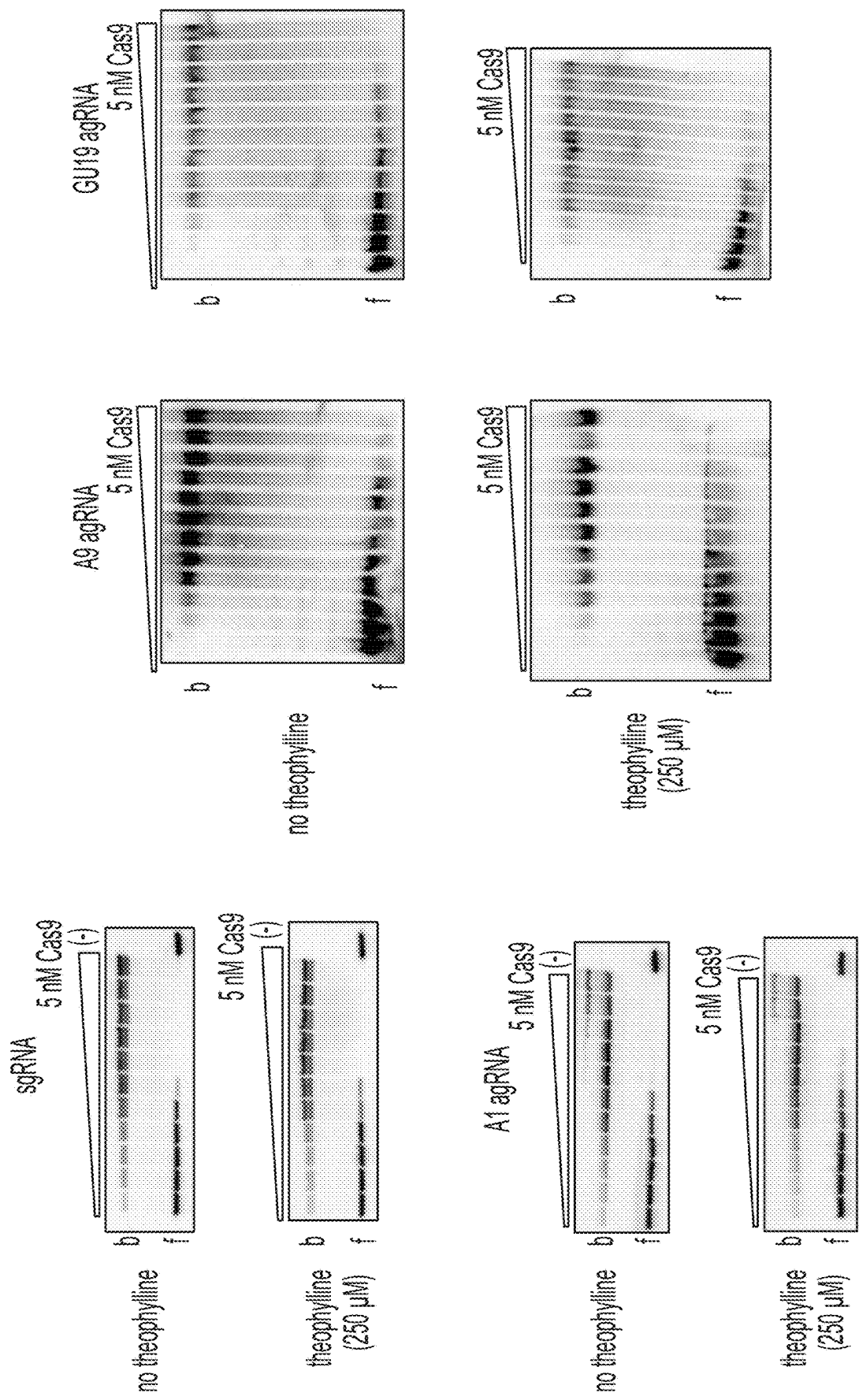


FIG. 10

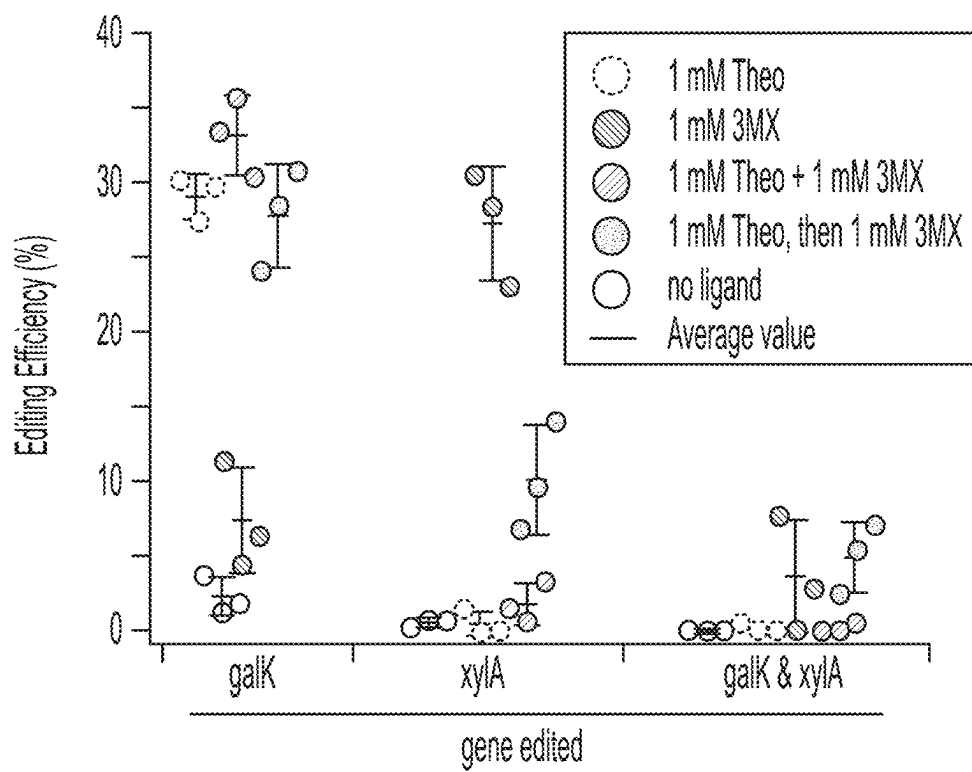


FIG. 11

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COMPOSITIONS AND METHODS OF USE FOR SMALL-MOLECULE REGULATION OF CRISPR-CAS9 ACTIVITY USING RNA APTAMERS

CROSS REFERENCE TO RELATED APPLICATIONS

This Continuation application claims priority to Patent Cooperation Treaty (PCT) Application No. PCT/US2020/013718, filed Jan. 15, 2020, which claims the benefit under 35 U.S.C. § 119(3) of the filing date of U.S. Provisional Application No. 62/793,748 filed Jan. 17, 2019. Each of these applications is hereby incorporated by reference in their entirety for all purposes.

GOVERNMENT LICENSE RIGHTS

This invention was made with government support under grant number GM073850 awarded by the National Institutes of Health. The government has certain rights in the invention.

SEQUENCE LISTING

The instant application contains a Sequence Listing which has been submitted electronically in ASCII format and is incorporated herein by reference in its entirety for all purposes. The ASCII copy, created on Oct. 28, 2024, is named 101877-696496 CU4364B-US1 Sequence_listing_ST25.txt and is 57,600 bytes in size.

BACKGROUND

CRISPR-Cas9 has led to great advances in gene editing for a broad spectrum of applications. To further the utility of Cas9, there have been efforts to achieve temporal control over its nuclease activity. While different approaches have focused on regulation of Cas9 or gRNA-regulated CRISPR interference, none of the reported methods enable stringent control of the nuclease activity in an orthogonal and multiplexed manner. As a result, there remains a need for improved methods to temporally control Cas9 activity.

SUMMARY

Provided herein are compositions and methods for temporal regulation of Cas enzyme activity, as well as methods of selecting compositions for the temporal regulation of Cas enzyme activity. Described herein is single guide RNA (sgRNA) that comprises a small-molecule-binding aptamer at a specific location, which enables small-molecule-dependent gene editing in bacteria. In some embodiments, the sgRNA comprises specific linking RNA sequences positioned between the sgRNA and the aptamer. Also described herein is a method for selecting sgRNAs that allow for small-molecule dependent gene editing in bacterium.

Accordingly, in some aspects, the disclosure provides a single guide RNA (sgRNA) comprising a small-molecule-binding aptamer sequence adjacent to a region comprising a 2×4 internal loop and upper stem.

In some embodiments, the small-molecule-binding aptamer is a theophylline-binding aptamer or 3-methylxanthine-binding aptamer. In some embodiments, the small-molecule-binding aptamer is a theophylline-binding aptamer and the theophylline-binding aptamer comprises the sequence 5'-AUACCAGCUUCGAAAGAAGCCC-UUGGCAG-3' (SEQ ID NO: 95). In some embodiments, the small-molecule-binding aptamer is a 3-methylxanthine-

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binding aptamer and the 3-methylxanthine-binding aptamer comprises the sequence 5'-AUACCAGC-UUCGAAAGAAGCCAUUGGCAG-3' (SEQ ID NO: 96).

In some embodiments, sgRNA contains 0-2 nucleotides between the small-molecule-binding aptamer sequence and the 2×4 internal loop and upper stem, optionally wherein the sgRNA contains 0 nucleotides between the small-molecule-binding aptamer sequence and the 2×4 internal loop and upper stem.

In some embodiments, the region comprising the 2×4 internal loop and upper stem comprises the sequence 5'-GNNNCGXCCUNNNNNNC-3' (SEQ ID NO: 168), 5'-GNNNGGXCCNNNNNNNC-3' (SEQ ID NO: 169), 5'-GNNNGGXCCNNNNNNNU-3' (SEQ ID NO: 170), 5'-GNNNGUXANNNNNNNNC-3' (SEQ ID NO: 171), 5'-ANNNGGXCCNNNNNNNU-3' (SEQ ID NO: 172), 5'-CNNNGGXANNNNNNNNG-3' (SEQ ID NO: 173), 5'-UNNNGGXNNNNNNA-3' (SEQ ID NO: 174), 5'-UNNNGGXNNNNNNNG-3' (SEQ ID NO: 175), 5'-UNNNNUXNNNNNNNU-3' (SEQ ID NO: 176) or 5'-GNNNGGXCCNNNNNNNU-3' (SEQ ID NO: 177), wherein N is any nucleotide and wherein X is the small-molecule-binding aptamer sequence.

In some embodiments, the region comprising the 2×4 internal loop and upper stem comprises any one of the sequences in Table 1, wherein X in Table 1 is the small-molecule-binding aptamer sequence:

TABLE 1

Clone Name	Sequence (5' To 3')	SEQ ID NO:
G1B1	GUAUCGXCUUAAGCC	97
RG53	AGUGAGXCUAAAAU	98
A38	GAGAGGXCCCCCGGC	99
A9	UGAAGGXCCGCAACA	100
A39	AGAAGGXCCCAUCAU	101
A34	UAGUUUXAAACCGUU	102
A1	GGGGGGXCCUAUUUU	103
A10	GAUGGGXCCUCCACC	104
A14	GGAGGUXACGGUGCC	105
A19	GAGAGGXCCCCCGGC	106
A8	CGGGGGXACAAUAGG	107
A26	GGGGGGXCCACGCGC	108
C2A2	UAGUGGXCUACCAUG	109
AU1	AGGGGGXCCUAUAAU	110
AU2	ACCAGGXCCAAGUAAU	111
AU3	ACAAGGXCCCAUAAU	112
AU5	ACUCGGXCCUGAACU	113
AU6	AAGGGGXCCUAUAAU	114
AU7	AGUAGGXCCUUUCAU	115
AU8	ACGGGGXCCUAUAAU	116
AU9	AAACGGXCCCACUGU	117
AU10	AUAGGGXCCAUCCAU	118

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TABLE 1-continued

Clone Name	Sequence (5' To 3')	SEQ ID NO:
AU11	AAUAGGXCCACUUAU	119
AU12	AGAGGGXCCGGGCGU	120
AU14	AGUGGGXCCAGCCUU	121
AU15	ACCCGGXCCAUAU	122
AU16	AACCGGXCCCCGAGU	123
AU17	AAAGGGXCCAGGCAU	124
AU18	AAUAGGXCCCAGACU	125
AU19	AAUAGGXCCCGCAGU	126
GU5	GGUUGGXCCUAUAU	127
GU6	GGACGGXCCAAGCAU	128
GU8	GGCAGGXCCUCUUCU	129
GU12	GCCGGGXCCUUUUUU	130
GU13	GGCCGGXCCAAGCAU	131
GU14	GACUGGXCCUAUAAU	132
GU15	GGACGGXCCUACAAU	133
GU18	GAUUGGXCCUACGGU	134
GU19	GAUCGGXCCAUAAGAU	135
GC3	GUCCGGXCCCCACAC	136
GC4	GGUCGGXCCAGUAGC	137
GC6	GAUUGGXCCAGCAAC	138
GC7	GGGGGGXCCGAUAC	139
GC11	GCGUGGXCCCUCUCC	140
GC12	GAUAGGXCCAGUUAC	141
GC13	GGAAGGXCCUUAUAC	142
GC15	GCAUGGXCCUACUCC	143
GC16	GAUAGGXCCAACACC	144
GC17	GACCGGXCCCCCGC	145
GC18	GAUUGGXCCGCAACC	146
GC20	GCAAGGXCCAACACC	147

In some embodiments, the region comprising the 2×4 internal loop and upper stem comprises any one of the sequences in Table 2, wherein X in Table 2 is the small-molecule-binding aptamer sequence:

TABLE 2

Clone Name	Sequence (5' To 3')	SEQ ID NO:
G1B1	GUAUCGXCUAAGCC	148
A38	GAGAGGXCCCCCGGC	149
A9	UGAAGGXCCGCAACA	150

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TABLE 2-continued

Clone Name	Sequence (5' To 3')	SEQ ID NO:
5 A34	UAGUUUXAAACCGUU	151
A1	GGGGGGXCCUAUUUU	152
A14	GGAGGUXACGGUGCC	153
10 A19	GAGAGGXCCCCCGGC	154
A8	CGGGGGXACAAUAGG	155
A26	GGGGGGXCCACGCGC	156
15 C2A2	UAGUGGXCUACCAUG	157
AU3	ACAAGGXCCCAUAAU	158
AU5	ACUCGGXCCUGAACU	159
20 AU6	AAGGGGXCCUAUAAU	160
AU7	AGUAGGXCCUUUCAU	161
AU8	ACGGGGXCCUAUAU	162
GU19	GAUCGGXCCAUAAGAU	163
25 GC7	GGGGGGXCCGAUAC	164
GC13	GGAAGGXCCUUAUAC	165
GC16	GAUAGGXCCAACACC	166
30 GC20	GCAAGGXCCAACACC	167

In some embodiments, the region comprising the 2×4 internal loop and upper stem comprises the sequence 5'-GGGGGGXCCUAUUUU-3'(SEQ ID NO: 103), 5'-UGAAGGXCCGCAACA-3' (SEQ ID NO: 100) or 5'-GAUCGGXCCAUAAGAU-3' (SEQ ID NO: 135).

In other aspects, the disclosure provides a ribonucleoprotein (RNP) comprising a sgRNA of any one of the above embodiments or as otherwise described herein and a Cas enzyme. In some embodiments, the Cas enzyme is Cas9.

In other aspects, the disclosure provides a method of inducing gene editing in a bacterium, the method comprising (a) introducing or expressing in a bacterium the sgRNA of any one of the above embodiments or as otherwise described herein, (b) introducing or expressing a Cas enzyme in the bacterium, and (c) contacting the bacterium with a small molecule that interacts with the small-molecule-binding aptamer sequence in the sgRNA such that gene editing is induced in the bacterium. In some embodiments, the small molecule is theophylline or 3-methylxanthine. In some embodiments, the method comprises introducing at least two sgRNAs into the bacterium, wherein a first sgRNA comprises a first small-molecule-binding aptamer sequence and a second sgRNA comprises a second small-molecule-binding aptamer sequence that is different than the first small-molecule-binding aptamer sequence, and wherein the method comprises contacting the bacterium with a first small molecule that interacts with the first small-molecule-binding aptamer sequence and contacting the bacterium with a second small molecule that interacts with the second small-molecule-binding aptamer sequence. In some embodiments, the bacterium is contacted with the first small molecule prior to the second small molecule, or vice versa. In some embodiments, the first small molecule is theophylline and the second small molecule is 3-methylxanthine. In some embodiments, the Cas enzyme is Cas9.

In yet other aspects, the disclosure provides an in vivo method of selecting a switchable aptamer single guide RNA (agRNA) that is inducible with a small molecule, the method comprising (a) performing a negative selection step comprising growing a first plurality of bacteria in the absence of the small molecule and in the absence of λ -red protein expression, thereby inducing cell death in a subset of the plurality to produce a second plurality of bacteria, wherein individual bacteria in the first plurality comprise a nucleic acid encoding a candidate agRNA that produces a cut site in the gene encoding a selection marker protein, a Cas enzyme, and a template for homologous repair of the cut site in the gene that encodes the selection marker protein; (b) performing a positive selection step comprising either (i) growing the second plurality of bacteria in the presence of the small molecule and λ -red protein expression or (ii) isolating nucleic acids encoding candidate agRNAs from the second plurality of bacteria and introducing the nucleic acids into a third plurality of bacteria and growing the third plurality of bacteria in the presence of the small molecule and λ -red protein expression, wherein individual bacteria in the third plurality comprise a nucleic acid encoding a candidate agRNA that produces a cut site in the gene encoding a selection marker protein, a Cas enzyme, and a template for homologous repair of the cut site in the gene that encodes the selection marker protein; and (c) selecting at least one bacterium from step (b) that expresses the altered selection marker protein. In some embodiments, the selection marker protein is galK and the template for homologous repair comprises a premature stop codon for the galK gene. In some embodiments, the bacteria is *E. coli*.

In yet other aspects, the disclosure provides in vivo method of selecting a switchable aptamer single guide RNA (agRNA) that is inducible with a small molecule, the method comprising (a) performing a negative selection step comprising growing a first plurality of bacteria in the absence of the small molecule and in the presence of λ -red protein expression, thereby inducing cell death in a subset of the plurality to produce a second plurality of bacteria, wherein individual bacteria in the first plurality comprise a nucleic acid encoding a candidate agRNA that produces a cut site in the gene encoding a selection marker protein, a Cas enzyme, and a template for homologous repair of the cut site in the gene that encodes the selection marker protein; (b) performing a positive selection step comprising either (i) growing the second plurality of bacteria in the presence of the small molecule and λ -red protein expression or (ii) isolating nucleic acids encoding candidate agRNAs from the second plurality of bacteria and introducing the nucleic acids into a third plurality of bacteria and growing the third plurality of bacteria in the presence of the small molecule and λ -red protein expression, wherein individual bacteria in the third plurality comprise a nucleic acid encoding a candidate agRNA that produces a cut site in the gene encoding a selection marker protein, a Cas enzyme, and a template for homologous repair of the cut site in the gene that encodes the selection marker protein; and (c) selecting at least one bacterium from step (b) that expresses the altered selection marker protein. In some embodiments, the selection marker protein is galK and the template for homologous repair comprises a premature stop codon for the galK gene. In some embodiments, the bacteria is *E. coli*.

BRIEF DESCRIPTION OF THE DRAWINGS

The following drawings form part of the present specification and are included to further demonstrate certain

aspects of the present disclosure, which can be better understood by reference to one or more of these drawings in combination with the detailed description of specific embodiments presented herein.

FIGS. 1A-1E show the design, selection and screening of agRNA libraries. FIG. 1A shows that the theophylline aptamer is inserted into the sgRNA at the site of the tetraloop used to fuse the guide and tracrRNAs. FIG. 1B shows that the 2x4 internal loop and flanking helices were randomized to yield agRNA libraries. FIG. 1C is a schematic showing that the switchable constructs were enriched in the two steps of the galK survival selection. FIG. 1D is an overview of constructs that were sampled from the enriched plasmid library after the galK survival selection. Editing was induced with 1 mM theophylline. The outer graph shows agRNAs are binned according to the difference in editing efficiency in an induced and uninduced population. The inset graph shows agRNAs are binned according to the fold increase in editing efficiency upon induction. FIG. 1E is a graph showing inducible gene editing was tested at sites other than the canonical galK1 site used for the galK selection (full circles, 1 mM theophylline added; empty circles, no theophylline added). The error bars represent the standard deviation from the mean.

FIGS. 2A-2C show the characterization of selected agRNAs. FIG. 2A is a heat map of the editing efficiency at the galK1 site with the agRNAs A9 and GU19 as a function of the concentration of theophylline and the time of recovery with theophylline after transformation. In FIGS. 2A-2B, the values represent the average of three biological replicates. FIG. 2B shows the editing efficiency at the galK1 site was measured for agRNAs composed of the theophylline or 3MX-aptamer coupled via the linker A9 or GU19 in dependence of 1 mM theophylline (Theo) or 1 mM 3MX. Recovery time was 3 hours. FIG. 2C shows the binding of Cas9 to sgRNA or agRNAs was observed and quantified via EMSA. Two representative binding curves for the A9 agRNA with and without 250 μ M theophylline are displayed. The inset table summarizes all measured dissociation constants. Error bars indicate standard deviation from the mean.

FIGS. 3A-3D show aptamer-gRNAs enable high-throughput genome editing. FIG. 3A is a schematic showing the workflow of the aptamer-controlled gene editing. In the left graph, FIG. 3B shows the transformation efficiencies of the MG λ 9 strain with wt gRNA and agRNAs that are activated with theophylline 1 hour after transformation. CFUs are colony forming units μ g-1 DNA and per 4 mL of recovery media. In the right graph, FIG. 3B shows the editing efficiencies with wt gRNA and ligand-activatable agRNAs that are activated with theophylline 1 hour after transformation. Each dot represents a biological replicate and error bars indicate standard deviation from the mean. FIG. 3C is a schematic of the plasmid design used to enable multiplexed gene editing. The individual homology arms and agRNAs are identical to the ones previously described for the single-edit experiments. FIG. 3D shows the editing efficiency at the two gene sites, galK (site 1) and xylA (site 2) in dependence on agRNA induction with either 1 mM theophylline and/or 1 mM 3MX. The values represent the average of three biological replicates.

FIGS. 4A-4B show the editing efficiencies at the galK 1 site. FIG. 4A is a graph showing that replacing the tetraloop in the wild-type sgRNA with the theophylline aptamer, but leaving the 2x4 loop unchanged, did not change the activity of the agRNA nor did it make the agRNA theophylline-dependent. agRNAs created by Liu et al.⁸, designed to regulate gene expression via CRISPRi, lead to gene editing

with and without the aptamer-ligand. Full circles: 1 mM theophylline added, empty circles: no theophylline added. FIG. 4B is a graph showing Cas9 expression was regulated by the pBAD promoter which is induced by addition of 0.2% arabinose. The number of transformants and the editing efficiency was independent of arabinose. Full circles: 0.2% (w/v) arabinose added, empty circles: no arabinose added. CFU: colony forming units. Every circle represents one biological replicate.

FIG. 5 is a schematic showing the work-flow of the screening process, based on the CREATE protocol. The bacteria were recovered for 3 hours after transformation with the aptamer ligand and then plated on MacConkey agar, containing galactose as the only source of sugar. White colonies were genomically unedited and unable to ferment galactose.

FIG. 6 shows the editing efficiencies of different agRNAs at the galK 1 site and the xylA site. Full circles: 1 mM theophylline. Empty circles: no theophylline.

FIG. 7 shows the traces from Sanger sequencing after colony PCR, confirming the genomic edit in colonies with white appearance and showing no edits in colonies with red appearance.

FIG. 8 is a pair of graphs which detail the single data points summarized in FIG. 2A. The underlying data is identical. Error bars indicate $\pm$ standard deviation from the mean.

FIG. 9 is a graph which details the single data points summarized in FIG. 2B. The underlying data is identical. Error bars indicate $\pm$ standard deviation from the mean.

FIG. 10 shows representative images of gels from the electrophoretic mobility shift assay. The upper band corresponds to P32-labeled agRNA bound by Cas9. The lower band corresponds to free P32-labeled agRNA. Images were taken on a Typhoon FLA 9500.

FIG. 11 is a graph which details the single data points summarized in FIG. 3D. The underlying data is identical. Error bars indicate $\pm$ standard deviation from the mean.

DETAILED DESCRIPTION

The utility of CRISPR for genome editing in bacteria such as *E. coli* has been demonstrated in various studies. However, CRISPR-based editing technologies in bacteria suffer from low transformation efficiencies caused by the lethality of double-stranded DNA (dsDNA) breaks in bacteria and from associated issues, such as biases in multiplexed libraries towards non-cutting gRNAs. As described herein, the tetraloop that fuses the crRNA and tracrRNA in a sgRNA was replaced with a small-molecule-binding aptamer sequence and, in some embodiments, modifications were made to the region that attaches the aptamer to the rest of the sgRNA, namely the 2 $\times$ 4 internal loop and upper stem. As described herein, gene editing is induced in bacteria expressing the modified sgRNAs by addition of a small molecule that interacts with the aptamer. Accordingly, the present disclosure provides compositions and methods for inducing gene editing in bacteria, e.g., in a temporally controlled manner.

Single Guide RNAs (sgRNAs)

In some aspects, the disclosure provides single guide RNAs (sgRNAs) that comprise an aptamer sequence, e.g., an aptamer sequence as described herein.

CRISPR (clustered regularly interspaced short palindromic repeats)/Cas is a prokaryotic antiviral system that has been repurposed for gene editing in a variety of cell types, including bacteria. In general, the CRISPR/Cas system relies

on the activity of two RNAs, a trans-activating CRISPR RNA (tracrRNA) and a CRISPR RNA (crRNA), which interact with a Cas enzyme (e.g., Cas9) and direct the Cas enzyme to cut the genome in a site-specific manner. The crRNA portion contains a protospacer, generally 20-nucleotides long, which determines the sequence specificity of the Cas nuclease activity. In general, the protospacer should be complementary to a region in the genome that is adjacent to a protospacer adjacent motif (PAM).

The sequence of the PAM depends on the type of Cas enzyme used. PAM sequences are known in the art and further described herein. The tracrRNA contains a stem loop structure that binds to the Cas enzyme. In their natural state, the tracrRNA and crRNA are separate molecules that bind together through a base pairing interaction. A single guide RNA (sgRNA) is a single synthetic molecule that comprises both a tracrRNA and a crRNA. sgRNAs are well-known in the art (see, e.g., Jinek et al. A programmable dual-RNA-guided DNA endonuclease in adaptive bacterial immunity. *Science*. 2012; 337(6096):816-21) and can be ordered through commercially available vendors (e.g., Dharmacon or New England Biolabs) or designed using the methods described in the Examples and/or through available software (e.g., CHOPCHOP v2: a web tool for the next generation of CRISPR genome engineering. *Nucleic Acids Res* 44(W1), W272-W276 (2016) and chopchop.cbu.uib.no). Generally, in a sgRNA the tracrRNA and crRNA are connected through a tetraloop sequence, which is connected to the rest of the sgRNA through a 2 $\times$ 4 internal loop and an upper stem. An example structure of an sgRNA showing the tetraloop, internal loop and upper stem is provided in FIG. 1A. As described herein, the tetraloop sequence may be replaced, in whole or in part, with an aptamer sequence, e.g., a small-molecule-binding aptamer sequence, and that aptamer sequence can be used to provide an inducible sgRNA. An example structure of an sgRNA showing the aptamer, internal loop and upper stem is provided in FIG. 1B.

Accordingly, in some embodiments, the disclosure provides an sgRNA comprising an aptamer sequence as described herein, e.g., a small-molecule-binding aptamer sequence. In some embodiments, the aptamer sequence, e.g., a small-molecule-binding aptamer sequence, replaces (in whole or in part) a tetraloop in the sgRNA. In some embodiments, the aptamer sequence, e.g., a small-molecule-binding aptamer sequence, is adjacent to (e.g., within 5, 4, 3, 2, 1 or 0 nucleotides of) a region of the sgRNA comprising a 2 $\times$ 4 internal loop and/or upper stem. In some embodiments, the aptamer sequence, e.g., a small-molecule-binding aptamer sequence, is located in a bridging region that joins the tracrRNA and crRNA regions of the sgRNA sequence.

In some embodiments, the region comprising the 2 $\times$ 4 internal loop and upper stem comprises the sequence 5'-NNNNNNXNNNNNNNN-3' (SEQ ID NO: 178), wherein N is any nucleotide and wherein X is an aptamer sequence (e.g., SEQ ID NO: 95 or 96 or 180). Typically, an aptamer sequence (e.g., as represented by X) is from about 10 nucleotides to about 50 nucleotides in length, e.g., from about 20 nucleotides to 40 nucleotides in length. In some embodiments, the region comprising the 2 $\times$ 4 internal loop and upper stem comprises the sequence 5'-GNNNCGXCUNNNNNNC-3' (SEQ ID NO: 168), 5'-GNNNGGXCCNNNNNNC-3' (SEQ ID NO: 169), 5'-GNNNGGXCCNNNNNNNU-3' (SEQ ID NO: 170), 5'-GNNNGGXANNNNNNNC-3' (SEQ ID NO: 171), 5'-ANNNGGXCCNNNNNNNU-3' (SEQ ID NO: 172), 5'-CNNNGGXANNNNNNNG-3' (SEQ ID NO: 173), 5'-UNNNGGXNNNNNA-3' (SEQ ID NO: 174), 5'-UN-

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NNGGXNNNNNG-3'(SEQ ID NO: 175), 5'-UNNN-UUXNNNNNU-3' (SEQ ID NO: 176) or 5'-GNNNGGXCCNNNNNU-3' (SEQ ID NO: 177), wherein N is any nucleotide and wherein X is an aptamer sequence, e.g., an aptamer sequence as described herein. In some embodiments, the region comprising the 2×4 internal loop and upper stem comprises any one of the sequences in in the Examples or in Table 1 or Table 2, wherein X in Table 1 and Table 2 is an aptamer sequence, e.g., an aptamer sequence as described herein. In some embodiments, the region comprising the 2×4 internal loop and upper stem comprises any one of the sequences in in the Examples or in Table 1 or Table 2 with up to 5, 4, 3, 2 or 1 substitutions, deletions, or additions. In some embodiments, the region comprising the 2×4 internal loop and upper stem comprises the sequence 5'-GGGGGGXCCUAUUUU-3' (SEQ ID NO: 103), 5'-UGAAGGXCCGCAACA-3' (SEQ ID NO: 100) or 5'-GAUCGGXCCAUAGAU-3' (SEQ ID NO: 135). In some embodiments, the region comprising the 2×4 internal loop and upper stem comprises the sequence 5'-GGGGGGXCCUAUUUU-3'(SEQ ID NO: 103), 5'-UGAAGGXCCGCAACA-3' (SEQ ID NO: 100) or 5'-GAUCGGXCCAUAGAU-3' (SEQ ID NO: 135) with up to 5, 4, 3, 2 or 1 substitutions, deletions, or additions.

TABLE 1

Example sequences		
Clone Name	Sequence (5' To 3')	SEQ ID NO:
G1B1	GUAUCGXCUUAAGCC	97
RG53	AGUGAGXCUGAAAAU	98
A38	GAGAGGXCCCCGGC	99
A9	UGAAGGXCCGCAACA	100
A39	AGAAGGXCCCAUCAU	101
A34	UAGUUUXAAACCGUU	102
A1	GGGGGGXCCUAUUUU	103
A10	GAUGGGXCCUCCACC	104
A14	GGAGGUXACGGUGCC	105
A19	GAGAGGXCCCCGGC	106
A8	CGGGGGXACAAUAGG	107
A26	GGGGGGXCCACGCGC	108
C2A2	UAGUGGXCUACCAUG	109
AU1	AGGGGGXCCUAUAAU	110
AU2	ACCAGGXCCAAGUUAU	111
AU3	ACAAGGXCCCAUAAU	112
AU5	ACUCGGXCCUGAACU	113
AU6	AAGGGGXCCUAUAAU	114
AU7	AGUAGGXCCUUUCAU	115
AU8	ACGGGGXCCUAUAAU	116
AU9	AAACGGXCCCACUGU	117
AU10	AUAGGGXCCAUCCAU	118

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TABLE 1-continued

Example sequences		
Clone Name	Sequence (5' To 3')	SEQ ID NO:
AU11	AAUAGGXCCACUUUAU	119
AU12	AGAGGGXCCGGGCGU	120
AU14	AGUGGGXCCAGCCUU	121
AU15	ACCCGGXCCAUUCAU	122
AU16	AACCGGXCCCCGAGU	123
AU17	AAAGGGXCCAGGCAU	124
AU18	AAUAGGXCCCAGACU	125
AU19	AAUAGGXCCCAGU	126
GU5	GGUUGGXCCUAAUUAU	127
GU6	GGACGGXCCAAGCAU	128
GU8	GGCAGGXCCUCUUCU	129
GU12	GCCGGGXCCUUUUUU	130
GU13	GGCCGGXCCAAGCAU	131
GU14	GACUGGXCCUAUAAU	132
GU15	GGACGGXCCUACAAU	133
GU18	GAUUGGXCCUACGGU	134
GU19	GAUCGGXCCAUAAGAU	135
GC3	GUCCGGXCCCCACAC	136
GC4	GGUCGGXCCAGUAGC	137
GC6	GAUUGGXCCAGCAAC	138
GC7	GGGGGGXCCGAAUAC	139
GC11	GCGUGGXCCCUUCCC	140
GC12	GAUAGGXCCAGUUAC	141
GC13	GGAAGGXCCUUUAC	142
GC15	GCAUGGXCCUACUCC	143
GC16	GAUAGGXCCAACACC	144
GC17	GACCGGXCCCCCGC	145
GC18	GAUUGGXCCGCAACC	146
GC20	GCAAGGXCCAACACC	147

TABLE 2

Further example sequences		
Clone Name	Sequence (5' To 3')	SEQ ID NO:
G1B1	GUAUCGXCUUAAGCC	148
A38	GAGAGGXCCCCGGC	149
A9	UGAAGGXCCGCAACA	150
A34	UAGUUUXAAACCGUU	151

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TABLE 2-continued

Further example sequences		
Clone Name	Sequence (5' To 3')	SEQ ID NO:
A1	GGGGGGXCCAUUUU	152
A14	GGAGGUXACGGUGCC	153
A19	GAGAGGXCCCCGGC	154
A8	CGGGGGXACAAUAGG	155
A26	GGGGGGXCCACGCGC	156
C2A2	UAGUGGXCUACCAUG	157
AU3	ACAAGGXCCCAUAAU	158
AU5	ACUCGGXCCUGAACU	159
AU6	AAGGGGXCCUAUAAU	160
AU7	AGUAGGXCCUUUCAU	161
AU8	ACGGGGXCCUAAUUAU	162
GU19	GAUCGGXCCAUGAU	163
GC7	GGGGGGXCCGAAUAC	164
GC13	GGAAGGXCCUUUUAAC	165
GC16	GAUAGGXCCAACACC	166
GC20	GCAAGGXCCAACACC	167

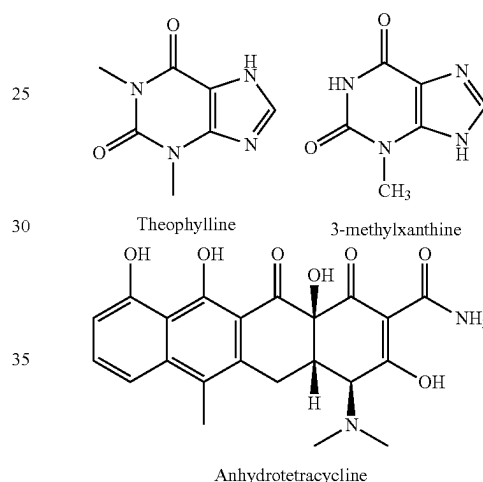
Aptamer

In some aspects, the disclosure provides an aptamer which is inserted into an sgRNA, e.g., an sgRNA as described herein. As used herein, an "aptamer" is an oligonucleotide molecule that binds to a target molecule. In some embodiments, the aptamer is an RNA aptamer that binds to a target molecule. In some embodiments, the target molecule is a small molecule, such that the aptamer is a small-molecule-binding aptamer. In some embodiments, the small-molecule-binding aptamer is an RNA aptamer that binds to a small molecule. In some embodiments, an aptamer (e.g., RNA aptamer) is up to about 50 nucleotides in length, e.g., up to 20 nucleotides, up to 30 nucleotides, or up to 40 nucleotides in length. In some embodiments, an aptamer (e.g., RNA aptamer) is at least 10 nucleotides in length, e.g., between 10 nucleotides to 50 nucleotides in length.

The small molecule can be any small molecule known in the art or described herein. Preferably, the small molecule is one that does not significantly negatively impact the viability of bacteria or significantly negatively impact gene editing in bacteria. In some embodiments, an aptamer can be designed and/or selected using methods known in the art (e.g., using SELEX (Systematic Evolution of Ligands by EXponential enrichment) or the methods described in Codrea et al. In Vitro Selection of RNA Aptamers to a Small Molecule Target. Curr. Protoc. Nucleic Acid Chem. (2010); 40:9.5.1-9.5.23) or through a commercial vendor (e.g., the Aptamer Group). In some embodiments, the small-molecule-binding aptamer binds to the small molecule with a dissociation constant (K_D) of no more than 1 micromolar, e.g., no more than 500 nanomolar, no more than 400 nanomolar, no more than 300 nanomolar, no more than 200 nanomolar or no more than 100 nanomolar.

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In some embodiments, the small-molecule-binding aptamer is a theophylline-binding aptamer or 3-methylxanthine-binding aptamer, optionally wherein the aptamer binds to theophylline or 3-methylxanthine, respectively, with a dissociation constant (K_D) of no more than 1 micromolar, e.g., no more than 500 nanomolar, no more than 400 nanomolar, no more than 300 nanomolar, no more than 200 nanomolar or no more than 100 nanomolar. In some embodiments, the small-molecule-binding aptamer is a theophylline-binding aptamer, a 3-methylxanthine-binding aptamer or an anhydrotetracycline-binding aptamer, optionally wherein the aptamer binds to theophylline, 3-methylxanthine or anhydrotetracycline, respectively, with a dissociation constant (K_D) of no more than 1 micromolar, e.g., no more than 500 nanomolar, no more than 400 nanomolar, no more than 300 nanomolar, no more than 200 nanomolar or no more than 100 nanomolar. The structures of theophylline, 3-methylxanthine, and anhydrotetracycline are shown below.



In some embodiments, the small-molecule-binding aptamer is a theophylline-binding aptamer and the theophylline-binding aptamer comprises the sequence 5'-AUACCAGCUUCGAAAGAAGCCCUUGGCAG-3' (SEQ ID NO: 95), optionally with up to 5, 4, 3, 2 or 1 substitutions, deletions, or additions. In some embodiments, the small-molecule-binding aptamer is a 3-methylxanthine-binding aptamer and the 3-methylxanthine-binding aptamer comprises the sequence 5'-AUACCAGCUUCGAAAGAAGCCAUUGGCAG-3' (SEQ ID NO: 96), optionally with up to 5, 4, 3, 2 or 1 substitutions, deletions, or additions. In some embodiments, the small-molecule-binding aptamer is an anhydrotetracycline-binding aptamer and the anhydrotetracycline-binding aptamer comprises the sequence 5'-GAGAGGUGAAGAAUACGACCACCUAGGUAGAAUACCUAAAACAUAC-3' (SEQ ID NO: 180), optionally with up to 5, 4, 3, 2 or 1 substitutions, deletions, or additions.

Ribonucleoproteins

Other aspects of the disclosure relate to a ribonucleoprotein (RNP) comprising an sgRNA, e.g., an sgRNA as described herein, and an RNA-guided endonuclease, such as a Cas enzyme. In some embodiments, the RNP may be expressed in or introduced into a bacterium. In some embodiments, a composition is provided that comprises an RNP, e.g., an RNP as described herein.

The RNA-guided endonuclease can be any RNA-guided endonuclease known in the art or described herein. In some embodiments, the RNA-guided endonuclease is a CRISPR class 2 type II Cas enzyme, including variants and homologs thereof. In some embodiments, the RNA-guided endonuclease is a Cas9 enzyme, including variants and homologs thereof. Examples Cas9 enzymes include *S. pyogenes* Cas9 (SpCas9), *S. aureus* Cas9 (SaCas9), *S. thermophilus* Cas9 (StCas9), *F. novicida* Cas9 (FnCas9), *N. meningitidis* Cas9 (NmCas9), and *B. laterosporus* Cas9 (BlatCas9), as well as variant forms of such enzymes (see, e.g., Nakade et al. Cas9, Cpf1 and C2c1/2/3—What's next? Bioengineered. 2017; 8(3): 265-273 and Komor et al. CRISPR-based technologies for the manipulation of eukaryotic genomes. Cell. 2017; 168(1-2): 20-36). PAM sequences corresponding to such enzymes are known in the art. Example PAM sequences for such enzymes include NRG, NGG, NGAG, NGCG, NGRRT, NGRRN, NNNNGATT, and NNAGAAW (see, e.g., Nakade et al. Cas9, Cpf1 and C2c1/2/3—What's next? Bioengineered. 2017; 8(3): 265-273 and International Patent Application Publication No. WO2016196805). In some embodiments, the Cas9 enzyme has at least 90% identity with (e.g., at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identity with) or comprises the following amino acid sequence (SEQ ID NO: 179):

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MDKKYSIGLDIGTNSVGWAVITDEYKVPSPKFKVLGNTDRHSIKKNLIG
ALLFDSGETAEATRLKRTARRRYTRRKNRICYLQEIFSNEMAKVDDSSFF
HRLEESFLVEEDKKHHERHPFIGNIVDEVAYHEKYPTIYHLRKKLV DSTD
KADLRLIYLALAHMIKFRGHFLIEGDLNPDNSDVKLFIQLVQTYNQLF
EENPINASGVDAKAII SARLSKSRRLLENLIAQLPGEEKNGLFGNLI ALS
LGLTPNFKSNFDLAEDAKLQLSKDTYDDDLNLLAQIGDQYADLFLAAK
NLSDAILLSDILRVNTEITKAPLSASMIKRYDEHHQDLTLKALVRQQL
PEKYKEIFPDQSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVK
LNREDLLRKQRTFDNGSIPHQIHLGELHAILRRQEDFYFPLKDNREKIE
KILTFRIPYVYVGLPARGNSRFAMWTRKSEETITPWNFEEVVDKGASAQ S
FIERMTNFDKNLPNEKVLPHKSLLEYFTVYNELTKVKYVTEGMRKPAF
LSGEQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEISGVEDRFN
ASLGTYHDL LKIKDKDFLDNEENEDILEDIVLTTLTFDREMIEERLK
TYAHLFDKKVMKQLKRRRYTGWGRLSRKLINGIRDKQSGKTILDFLKSD
GFANRNFMQLIHDDSLTFKEDIQKAQVSGQGDSLHEHIANLAGSPA I KK
GILQTVKVVDELVKVMGRHKPENIVEMARENQTQKGQKNSRERMKRI
EEGIELGSGQILKEHPVENTQLQNEKLYLYLQNGRDMYVDQELDINRL
SDYDVHIVPQSF LKDDSIDNKVLRSDKNRGSNDNVPSEEVVKMKMNY
WRQLLNAKLI TQRKFDNLTKAERGLSELDKAGFIKRLVETRQITKHV
AQILDSRMNTKYDENDKLIREVKVITLKSCLVSDFRKDFQFYKVR EINN
YHHAHDAYLNAVGTALIKKYPKLESEFVYGDYKVDVRKMI AKSEQEI
GKATAKYFFYSNMNFEKTEITLANGEIRKRLIETNGETGEI VWDKGR
DFATVRKVLSPMPQVNI VKKTEVQTGGFSKESILPKRNSDKLIARKKDW D
PKKYGGFDSPTVAYSVLVAKVEKGSKKLKSVKELLGITIMERS SFEK
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NPIDFLEAKGYKEVKKDLIIKLPKYSLFELENGRKRMLASAGELQK GNE
LALPSKYVNFYLYLASHYEKLKGSPEDEQKQLFVEQHKHYLDEIIEQIS
EFSKRVILADANLDKVL SAYNKHDKPIREQAENIIHLFTLTNLGAPAA
FKYFDTTIDRKRYTSTKEVL DATLIHQSI TGLYETRIDLSQLGGD
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Methods of Inducing Gene Editing

Yet other aspects of the disclosure relate to a method of inducing gene editing in a bacterium. As used herein, "inducing" includes both (a) causing a bacterium in which no editing is occurring to become a bacterium in which gene editing occurs to some extent and (b) increasing, in a bacterium in which at least some editing is occurring, the level of editing occurring (e.g., increasing the editing efficiency in a given bacterium from 1% editing efficiency to 50% editing efficiency). In both (a) and (b), the extent to which gene editing is occurring is increased (in (a), from 0 to some and in (b), from some to more).

In some embodiments, the method comprises (a) introducing or expressing an sgRNA, e.g., an sgRNA as described herein (e.g., comprising a small-molecule-binding aptamer sequence adjacent to a region comprising a 2×4 internal loop and upper stem), in a bacterium, (b) introducing or expressing a Cas enzyme, e.g., a Cas enzyme as described herein (e.g., a Cas9 enzyme), in the bacterium, and (c) contacting the bacterium with a small molecule, e.g., a small molecule as described herein, that interacts with the aptamer sequence, e.g., an aptamer sequence as described herein (e.g., small-molecule-binding aptamer sequence), in the sgRNA such that gene editing is induced in the bacterium. In some embodiments, step (c) is performed after steps (a) and (b), e.g., at least 5 minutes, at least 10 minutes, at least 20 minutes, at least 30 minutes, at least 50 minutes, at least 1 hour, at least 2 hours, at least 4 hours, at least 8 hours, at least 12 hours, at least 24 hours after steps (a) and (b). In some embodiments, steps (a) and (b) are performed sequentially. In some embodiments, steps (a) and (b) are performed at the same time. In some embodiments, steps (a), (b) and (c) are performed at the same time.

Each of the sgRNA and the Cas enzyme independently may either be introduced, e.g., by permeabilizing the cell, or may be expressed, e.g., in using a plasmid or other vector that is present in the bacterium. In some embodiments, both the sgRNA and Cas enzymes are expressed, e.g., in using a plasmid or other vector that is present in the bacterium. Example plasmids for expressing a Cas enzyme and an sgRNA are described in the Examples.

The small molecule can be any small molecule described herein. In some embodiments, the small molecule is theophylline or 3-methylxanthine. In some embodiments, the small molecule is contacted with the bacterium at a concentration of at least 0.001 mM, at least 0.01 mM, at least 0.1 mM, at least 1 mM, at least 10 mM or at least 100 mM or more. In some embodiments, the small molecule is contacted with the bacterium at a concentration of between 0.001 mM to 100 mM, e.g., 1 mM.

In some embodiments, the method comprises introducing at least two sgRNAs into the bacterium, wherein a first sgRNA comprises a first small-molecule-binding aptamer sequence, e.g., a small-molecule-binding aptamer sequence as described herein, and a second sgRNA comprises a second small-molecule-binding aptamer sequence, e.g., a small-molecule-binding aptamer sequence as described herein, that is different than the first small-molecule-binding

aptamer sequence. In some embodiments, the first sgRNA targets a different region in the bacterial genome for editing than the second sgRNA (e.g., the first sgRNA and the second sgRNA contain different protospacer sequences in the crRNA region). In some embodiments of this method, the method further comprises contacting the bacterium with a first small molecule that interacts with the first small-molecule-binding aptamer sequence and contacting the bacterium with a second small molecule that interacts with the second small-molecule-binding aptamer sequence. In some embodiments, the first small molecule is theophylline and the second small molecule is 3-methylxanthine. The first small molecule and second small molecule may be contacted with the bacterium at different times or at the same time. In some embodiments, the bacterium is contacted with the first small molecule prior to the second small molecule. In some embodiments, the bacterium is contacted with the second small molecule prior to the first small molecule. In some embodiments, the time between contact with the first small molecule and second small molecule is at least 1 hour, at least 2 hours, at least 3 hours, at least 4 hours, or more. In some embodiments, the time between contact with the first small molecule and second small molecule is between 1 to 10 hours, 1 to 5 hours, 1 to 3 hours, 2 to 10 hours, 3 to 10 hours, 2 to 5 hours, or 3 to 5 hours. In some embodiments, having a time delay between contact with the first small molecule and the second small molecule allows for sequential editing, which may result in improved editing efficiency.

The bacterium used in the method may be any bacterium known in the art or described herein. In some embodiments, the bacterium is a Gram-negative bacterium. In some embodiments, the bacterium is an *E. coli* bacterium.

Methods of Selection

Other aspects of the disclosure relate to methods for selecting one or more switchable aptamer single guide RNAs (agRNAs) that are inducible with a small molecule and/or screening candidate agRNAs. As described herein, in some embodiments, a sgRNA containing an aptamer may need to be optimized or screened for desired inducibility with a small molecule. In some embodiments, the region comprising the 2x4 internal loop and/or upper stem of a sgRNA may need to be optimized or screened such that the aptamer adjacent to the region has a desired inducibility with a small molecule.

As described herein, agRNAs can be selected by a multi-step process. In some embodiments, the method comprises a negative selection step to eliminate candidate agRNAs that are constitutively active followed by a positive selection step to select candidate agRNAs that can be induced with the small molecule to edit a location in the genome. In some embodiments, the negative selection step involves growing the bacteria under conditions such that constitutively active agRNAs will induce double-stranded breaks, which will cause cell death. An example of such conditions includes growing the bacteria in the absence of the small molecule and in the absence of λ -red proteins, which normally increase the frequency of homologous repair of the double-stranded breaks. Candidate agRNAs that do not induce cell death are then subjected to the positive selection step. In some embodiments, the positive selection step involves growing the bacteria under conditions such that bacteria containing inducible agRNAs can be positively identified, e.g., using a selection marker that is identifiable in bacteria containing inducible agRNAs. An example of such conditions includes using galK selection, where an agRNA that is inducible facilitates the introduction of a stop codon into the galK gene such that a bacterium containing the mutated

galK gene will survive incubation with 2-deoxygalactose because such bacterium will not metabolize 2-deoxygalactose. In other embodiments, the negative selection step may comprise a galK selection step that comprises growing the bacteria in the absence of the small molecule and in the presence of λ -red protein. Active agRNAs will introduce a stop codon into the galK gene and the selection is carried out in a minimal media that contains galactose as the only carbon source, such that only bacteria containing inactive agRNAs will have a functional galK gene and will survive in the minimal media. The surviving bacteria may then be subjected to the positive selection step. In some embodiments, the negative selection comprises cell death by dsDNA cleavage in the absence of λ -red proteins and the positive selection step comprises survival by introducing a survival-conferring mutation through homologous recombination assisted by λ -red proteins.

In some embodiments, the method comprises (a) performing a negative selection step comprising growing a first plurality of bacteria in the absence of the small molecule and in the absence of λ -red protein expression, thereby inducing cell death in a subset of the plurality to produce a second plurality of bacteria, wherein individual bacteria in the first plurality comprise (i) a nucleic acid encoding a candidate agRNA that produces a cut site in the gene encoding a selection marker protein, (ii) a Cas enzyme (e.g., Cas9), and (iii) a template for homologous repair of the cut site in the gene that encodes the selection marker protein; (b) performing a positive selection step comprising either (i) growing the second plurality of bacteria in the presence of the small molecule and λ -red protein expression; or (ii) isolating nucleic acids encoding candidate agRNAs from the second plurality of bacteria and introducing the nucleic acids into a third plurality of bacteria and growing the third plurality of bacteria in the presence of the small molecule and λ -red protein expression, wherein individual bacteria in the third plurality comprise (i) a nucleic acid encoding a candidate agRNA that produces a cut site in the gene encoding a selection marker protein, (ii) a Cas enzyme, and (iii) a template for homologous repair of the cut site in the gene that encodes the selection marker protein; and (c) selecting at least one bacterium from step (b) that expresses the altered selection marker protein. In some embodiments, the steps of the method may be iterated, e.g., steps (a), (b) and/or (c) may be repeated once, twice, three times, four times, or more.

In other embodiments, the method comprises (a) performing a negative selection step comprising growing a first plurality of bacteria in the absence of the small molecule and in the presence of λ -red protein expression, thereby inducing cell death in a subset of the plurality to produce a second plurality of bacteria, wherein individual bacteria in the first plurality comprise (i) a nucleic acid encoding a candidate agRNA that produces a cut site in the gene encoding a selection marker protein, (ii) a Cas enzyme (e.g., Cas9), and (iii) a template for homologous repair of the cut site in the gene that encodes the selection marker protein; (b) performing a positive selection step comprising either (i) growing the second plurality of bacteria in the presence of the small molecule and λ -red protein expression; or (ii) isolating nucleic acids encoding candidate agRNAs from the second plurality of bacteria and introducing the nucleic acids into a third plurality of bacteria and growing the third plurality of bacteria in the presence of the small molecule and λ -red protein expression, wherein individual bacteria in the third plurality comprise (i) a nucleic acid encoding a candidate agRNA that produces a cut site in the gene encoding a selection marker protein, (ii) a Cas enzyme, and (iii) a

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template for homologous repair of the cut site in the gene that encodes the selection marker protein; and (c) selecting at least one bacterium from step (b) that expresses the altered selection marker protein. In some embodiments, the steps of the method may be iterated, e.g., steps (a), (b) and/or (c) may be repeated once, twice, three times, four times, or more.

In some embodiments, the candidate agRNA(s) is/are sgRNA(s), e.g., sgRNA(s) as described herein, e.g., comprising a small-molecule-binding aptamer sequence adjacent to a region comprising a 2×4 internal loop and upper stem. In some embodiments, the nucleic acid encoding the candidate agRNA is a plasmid. In some embodiments, the template for homologous repair is contained within a plasmid. In some embodiments, the nucleic acid encoding the candidate agRNA and the template for homologous repair are contained within the same plasmid. In some embodiments, bacteria in the first and/or second and/or third plurality comprise a plasmid that encodes a Cas enzyme (e.g., Cas9) and a plasmid that encodes a λ -red protein. In some embodiments, bacteria in the first and/or second and/or third plurality comprise a plasmid that encodes a Cas enzyme (e.g., Cas9) and that encodes a λ -red protein. Example plasmids that encode Cas9 and λ -red protein include X2-Cas9 (Addgene plasmid number #85811) and pSIM5 (see, e.g., Datta et al. A set of recombineering plasmids for gram-negative bacteria. *Gene*. (2006); 379, 109-115).

The selection marker protein can be any selection marker protein that allows for positive identification of inducible agRNAs. In some embodiments, the selection marker protein is an antibiotic resistance protein, a metabolic protein that enable survival in a defined media or an engineered kill-switch. In some embodiments, the selection marker protein is galK and the template for homologous repair comprises a premature stop codon for the galK gene. In some embodiments, when the selection marker protein is galK, then step (a) of the method is performed in a minimal media that contains galactose as the only carbon source. In some embodiments, when the selection marker protein is galK, then step (B) of the method is performed in a media comprising 2-deoxygalactose. In some embodiments, the selection marker protein is two proteins, e.g., a tetA protein and a sacB protein (see, e.g., Li et al. Positive and negative selection using the tetA-sacB cassette: recombineering and P1 transduction in *Escherichia coli*, *Nucleic Acids Res.* 2013 December; 41(22):e204).

The small molecule can be any small molecule described herein. In some embodiments, the small molecule is theophylline or 3-methylxanthine. In some embodiments, the small molecule is contacted with the plurality of bacteria at a concentration of at least 0.1 mM, at least 1 mM, at least 10 mM or at least 100 mM or more. In some embodiments, the small molecule is contacted with the plurality of bacteria at a concentration of between 0.1 mM to 100 mM, e.g., 1 mM.

In some embodiments, the first and/or second and/or third plurality of bacteria comprises at least 10^1 , at least 10^2 , at least 10^3 , at least 10^4 , at least 10^5 , at least 10^6 , at least 10^7 , at least 10^8 , at least 10^9 , or at least 10^{10} bacteria. The bacteria used in the method may be any bacteria known in the art or described herein. In some embodiments, the bacteria is gram-negative bacteria. In some embodiments, the bacteria is *E. coli* bacteria.

Without further elaboration, it is believed that one skilled in the art can, based on the above description, utilize the present disclosure to its fullest extent. The following specific embodiments are, therefore, to be construed as merely illustrative, and not limitative of the remainder of the disclosure in any way whatsoever. All publications cited

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herein are incorporated by reference for the purposes or subject matter referenced herein.

EXAMPLES

Example 1

CRISPR-Cas9 has led to great advances in gene editing for a broad spectrum of applications. To further the utility of Cas9, there have been efforts to achieve temporal control over its nuclease activity. While different approaches have focused on regulation of Cas9 or gRNA-regulated CRISPR interference, none of the reported methods enable stringent control of the nuclease activity in an orthogonal and multiplexed manner. Here, novel RNA linkers were developed to combine theophylline- and 3-methylxanthine (3MX)-binding aptamers with the gRNA, enabling small molecule-dependent editing in *Escherichia coli*. These activatable guide RNAs enable orthogonal, temporal and post-transcriptional control of in vivo gene editing. Further, they reduce the death of host cells caused by cuts in the genome, a major limitation of CRISPR-mediated bacterial recombineering. Temporal control of the enzyme will simultaneously increase library coverage and decrease bias in multiplex CRISPR/Cas9 editing experiments.

The utility of CRISPR for genome editing in *E. coli* has been demonstrated in various studies. One of those newly developed methods, called CREATE, uses plasmid-based recombineering, as opposed to the use of oligos, which enables easy tracking of the mutations in a library. However, CREATE, like other CRISPR-based editing technologies, suffers from low transformation efficiencies caused by the lethality of dsDNA breaks in bacteria and from associated issues, such as biases in multiplexed libraries towards non-cutting gRNAs. For this reason, inducible gRNAs were developed that would allow timing and titration of the nuclease activity to alleviate these issues. Previously, inducible gRNAs were developed for CRISPRi applications but could not be applied for inducible gene editing (FIG. 4A) due to the associated “leakiness”—the same reason why inducible promoters could not be used to solve the problem (FIG. 4A).

CREATE technology was used as a platform to develop ligand-switchable sgRNAs to reduce the DNA damage stress on the host bacteria. As a first step, the tetraloop used to fuse the crRNA and tracrRNA¹⁶ (FIG. 1A) was replaced with an in vitro selected small molecule binding aptamer (FIG. 1A). This site is highly tolerant of insertions¹⁷ and a theophylline aptamer-sgRNA fusion with an unchanged internal loop (IL) is constitutively active (FIG. 4A). To generate switchable aptamer-sgRNAs or “agRNAs”, a region including the IL and a small helix (upper stem) that connects the aptamer with the sgRNA was randomized. Nucleotides in the IL are critical for nuclease activity¹⁸ and directly interact with the PAM-interacting (PI) domain of Cas9¹⁹. Without wishing to be bound by theory, it is hypothesized that specific sequences could communicate ligand-dependent conformational changes in the aptamer to Cas9 to regulate nuclease activity (FIG. 1).

A fully randomized 14 nucleotide (14N) library was subjected to an in vivo survival assay in *E. coli*. Each agRNA of the library was cloned into a plasmid vector constitutively expressing the agRNA and containing the template for homologous repair of the cut site targeted by the agRNA. The cloned plasmid library was transformed into *E. coli* MG1655 to enrich theophylline-dependent agRNAs using a galK selection assay²⁰ in liquid culture. The

MG1655 strain used for the selection also carries the pSIM521 plasmid which expresses the λ -red proteins from a heat-inducible promoter and pX2-Cas922 which expresses Cas9 from the arabinose-inducible pBAD promoter. This strain will be referred to as MG λ 9. In the first selection step MG λ 9 was transformed with the agRNA library but theophylline was not added to the recovery media so switchable agRNAs in the library would be inactive. This step was performed without inducing expression of the λ -red proteins but with induced Cas9 expression so constitutively active agRNAs that target the galK1 site in the galK gene generate a double-stranded DNA break leading to cell death, eliminating constitutively active agRNA constructs from the selection (FIG. 1C). The second selection step was facilitated by the Cas9-mediated recombineering technology CREATE[®]. First, the MG λ 9 strain was transformed with the recovered plasmid library from the first step and theophylline was added to the recovery media to induce switchable agRNAs. Heat-induced expression of λ -red proteins enabled repair of the DNA break at the site galK1, created by activated agRNAs. Homologous recombination using a template provided on the agRNA plasmid introduced a premature stop codon into the galK gene preventing them from fermenting galactose. Edited cells could then be selected for by growing in minimal media containing the toxic galactose analog 2-deoxygalactose; only recombined bacteria that do not metabolize 2-deoxygalactose survive. Therefore, bacteria expressing activatable agRNAs in the presence of theophylline are enriched in this step (FIG. 1C). These selection/counterselection steps were iterated three times.

Candidates from this enriched library were analyzed to inform an improved library design. After three selection/counterselection steps, cells were plated on M63 selection medium²⁰ and 150 colonies were picked. Each isolated candidate was then transformed into unedited *E. coli* to screen for theophylline-responsive constructs using a red/white colony assay to quantify the percentage of edited bacteria. Colonies with an unedited galK site appear red whereas edited colonies appear white (FIG. 5). The percentage of white colonies is considered the editing efficiency. Gene editing was induced as previously described with or without adding theophylline and the editing efficiencies were compared.

Analysis of the 150 colonies from the 14N theophylline agRNA library after the selection yielded 16 theophylline-responsive constructs (Table 3), of which 3 constructs (A1, A9, A14) were found twice. In these sequences, a strong preference for Watson-Crick base pairs was observed in the regions flanking the IL (Table 4). Based upon this insight, more restricted libraries were generated in which three base pairs were fixed, which were predicted to contain a higher percentage of switchable agRNAs within a lower total number of sequences. Three 8N libraries ($\sim 6.6 \times 10^4$ sequences) were created that included two G-C pairs in the upper stem and one base pair at the 3'-side of the repeat/antirepeat helix (FIG. 1). Both libraries were transformed in sufficient numbers so as to ensure complete library coverage with 95% confidence²⁴. After the galK selection assay was used to enrich switchable constructs from those libraries, 20 colonies were screened from each library. Three sequences (GC7, GC10, GU10) out of the 60 from the three libraries were found twice. The low redundancy among the sequences of the initial and optimized libraries shows that even after enrichment, the screening only covered a fraction of the switchable constructs. The editing efficiencies of the individual constructs from these libraries are displayed in Table 3. Although most screened agRNA constructs were consti-

tutively inactive (FIG. 1D), 13 out of 60 constructs from the restricted theophylline libraries showed a >50-fold increase in editing efficiency (FIG. 1D) as opposed to 5 out of 150 from the initial screen. This demonstrates that the optimization of the library design was successful in increasing the fraction of switchable constructs in the library.

Promising agRNA constructs were analyzed in further detail. 17 agRNA constructs showed a combined >10-fold induction and >40% editing efficiency when induced and were selected for further screening. The xylA gene was targeted with these 17 agRNAs to test whether a different spacer sequence affects the performance. Editing was quantified with the same red/white screen. Two agRNAs (A9 and GU19) showed an editing efficiency at xylA similar to the galK1 site used during selection, while 9 others showed >10% editing at xylA (FIG. 6). To confirm sequence independence of those two agRNAs, they were used to edit three more alternative sites at the galK gene. Only one of those sites was not targetable with agRNA A9; GU19 was more site-dependent, but also showed less background activity (FIG. 1E). Sequencing confirmed that the phenotypic change in the assay corresponded to a designed genomic edit (FIG. 7). The observation that a number of agRNAs only act at the selected galK1 site highlights that the CRISPR-Cas9 system can be selected to act at a single sequence, a feature that could be exploited for reducing off-target effects.

To investigate the effect of ligand binding on the gene editing, the concentration and induction time with theophylline was systematically varied. This showed that the editing efficiency increased with the concentration of theophylline and induction time (FIG. 2A, FIG. 8). The concentration of theophylline in the media needed for rapid editing greatly exceeds the K_D of the isolated aptamer (400 nM), an observation typical for synthetic or natural ligand-activatable RNA devices. To test whether the binding of the ligand to the aptamer is the trigger of the agRNA activation, the C22A point mutation was introduced at the ligand-binding site of the theophylline aptamer domain²⁸. As expected, this turned the theophylline aptamer into a 3-methylxanthine aptamer, that was not activatable by theophylline (FIG. 2B, FIG. 9). This not only illustrated that ligand recognition by the aptamer is necessary for agRNA activation, it also expands the toolbox by an orthogonal agRNA.

To understand the effects of ligand binding to the aptamer on the activity of Cas9, the binding of the agRNA to the Cas9 protein was investigated. Binding of ³²P-labeled agRNA to Cas9 was quantified using an electrophoretic mobility shift assay (EMSA) (FIG. 10) in the presence or absence of 250 μ M theophylline. A1, A9 and GU19 agRNAs all exhibit the same apparent binding affinity to the Cas9 protein as the wild type gRNA, independent of theophylline (FIG. 2C). This indicated that despite the expected disruption of protein-RNA contacts in the randomized region, the RNP assembly is not ligand-dependent. Instead, without wishing to be bound by theory, it is hypothesized that the ligand regulates the recognition and/or cleavage of the DNA target.

Equipped with temporally controllable agRNAs, the gene editing technology CREATE¹⁰ was optimized. One major drawback of CRISPR-mediated bacterial gene editing technologies is cell death caused by dsDNA cuts¹⁰, despite overexpression of λ -red proteins to increase the frequency of homologous repair. Without wishing to be bound by theory, it is hypothesized that the stress of transformation combined with rapid generation of dsDNA breaks synergistically leads to a high level of cell death. Using the A9 or GU19 agRNAs

resulted in a drastic 10^4 -fold increase in number of transformants while maintaining ~80% editing efficiencies (FIG. 3B). Although a low transformation efficiency is not a significant issue when creating a single edit with a defined sgRNA, it dramatically impacts the ability to transform a library of sgRNA plasmids to generate a population of bacteria with different edits. In this process, a portion of gRNAs from the library is expected to be non-functional due to errors in oligonucleotide synthesis, sgRNA misfolding, or inefficient targeting.

Transformants with non-functional gRNAs do not suffer Cas9-mediated DNA cuts and have a strong fitness advantage over transformants with functional gRNAs, resulting in a population dominated by wild type cells. To simulate the transformation of a library containing non-functional gRNAs, bacteria were transformed with a plasmid mixture in which 90% encoded wild type gRNAs that target the galK1 site and are therefore functional and 10% encoded wild type gRNAs that target a site in the eGFP gene, which is absent in this strain, and are therefore non-targeting. Under these conditions, the percentage of edited cells dropped from 89% to 14% (FIG. 3D) and it was sequence-verified that most of the unedited cells carried plasmids expressing non-targeting gRNAs (Table 5). This corresponds to an 8.6 ± 0.1 -fold enrichment of the non-targeting plasmids. In contrast, using agRNAs A9 and GU19, the percentage of edited cells was maintained at 81% and 63%, respectively, which shows that non-targeting plasmids were not enriched in the process. This result indicates that separating the stresses caused by transformation and double-stranded breaks can dramatically increase cell survival, hence decreasing the fitness differences between cells harboring functional and non-functional gRNAs.

To further improve the utility of CRISPR-mediated recombineering, the agRNAs were applied to the problem of combinatorial edits. A hurdle impeding rapid throughput of combinatorial and multiple edits is that *E. coli* does not typically survive two simultaneous dsDNA breaks, even with induction of the λ -red proteins to facilitate homologous repair¹⁰. Therefore, only one mutation can be introduced in one round of CREATE editing. It was envisioned that the agRNAs would allow precise control over the timing of each

editing event, facilitating the introduction of multiple designed edits into the same cell. A plasmid was designed that expressed a theophylline-regulated GU19-agRNA targeting galK1 and a 3-methylxanthine (3MX)-regulated A9-agRNA targeting the xylA gene. Furthermore, the plasmid also contained the homology arms necessary for repair and mutation of the respective cut sites (FIG. 3C). The plasmid construct was then used for a 3-hour induction with either 1 mM theophylline, 1 mM 3MX or a combination of the two and the transformed bacteria were plated on MacConkey agar that contained either galactose, xylose or both to assay editing of the galK and xylA genes with the red/white screen (FIG. 3D, FIG. 11). As expected, adding theophylline induced editing of the galK gene but not the xylA gene and adding 3MX induced mostly editing of xylA and to a lesser extent galK. This corresponds to previously described promiscuity of the theophylline aptamer²⁹. Interestingly, a small number of transformants carried both mutations as a result. Simultaneously adding theophylline and 3MX led mostly to galK editing and few xylA edits. Adding first theophylline and after 3 hours adding 3MX for 2 hours led to an improved editing efficiency of xylA and resulted in transformants carrying both genomic edits. Thus, agRNAs can achieve CRISPR-mediated multiple genome editing in bacteria in a single step on a scale that allows the coverage of commonly used libraries.

In summary, an in vivo selection was developed which can be used to enrich ligand-activatable gRNAs. By stringently selecting for switchable agRNAs with low background activity, agRNAs that overcome the issue of leakiness associated with inducible promoters were obtained and allowed for a drastic improvement in the throughput of CRISPR-mediated recombineering by increasing the transformation efficiency 10^4 -fold and eliminating the bias for non-functional gRNAs. Furthermore, the combination of different guide-sequences with different aptamer-domains that bind distinct ligands allows orthogonal multiplexing of the agRNA activities enabling multiple, targeted mutations in a single experiment. The observation that many selected agRNAs could not recognize target sites other than the DNA target used in the selection might also show a way forward to reduce off-target cleavage in CRISPR applications.

TABLE 3

Name	Sequence	Editing Efficiency				SEQ ID
		no Theo	$\pm$ SD*	1 mM Theo	$\pm$ SD*	
sgRNA	AGAGCTAGCAAGTT	85%***	1%	89%***	3%	1
Cassettes from initial Library						
G1B1	GTATCGCTTAAGCC	1%		59%		2
RG53	AGTGAGCTAAAAAT	0%		22%		3
A38	GAGAGGCCCCCGGC	5%		85%		4
A9	TGAAGGCCGCAACA	9%***	7%	81%***	7%	5
A39	AGAAGGCCCATCAT	0%		7%		6
A34	TAGTTTAAACCGTT	12%		57%		7
A1	GGGGGGCCTATTTT	0%		66%		8
A10	GATGGGCCTCCACC	0%		14%		9
A14	GGAGGTACGGTGCC	2%		87%		10
A19	GAGAGGCCCCCGGC	20%		90%		11

TABLE 3-continued

Name	Sequence	Editing Efficiency				SEQ ID
		no Theo	±SD*	1 mM Theo	±SD*	
A8	CGGGGGACAATAGG	2%		82%		12
A26	GGGGGGCCACGCGC	5%		89%		13
C2A2	TAGTGGCTACCATG	5%		84%		14
Cassettes from optimized libraries						
AU1	AGGGGGCCTATAAT	0%		2%		15
AU2	ACCAGGCCAAGTAT	0%		4%		16
AU3	ACAAGGCCCATAAT	1%		72%		17
AU4	AACAGGCCCTAATT	0%		0%		18
AU5	ACTCGGCCTGAACT	0%		43%		19
AU6	AAGGGGCCTATAAT	5%		77%		20
AU7	AGTAGGCCTTTCAT	0%		58%		21
AU8	ACGGGGCCTAATAT	0.3%		48%		22
AU9	AAACGGCCCACTGT	1%		22%		23
AU10	ATAGGGCCATCCAT	0%		13%		24
AU11	AATAGGCCACTTAT	0%		25%		25
AU12	AGAGGGCCGGGCGT	0%		26%		26
AU14	AGTGGGCCAGCCTT	0%		17%		27
AU15	ACCCGGCCATTCT	0%		5%		28
AU16	AACCGGCCCCGAGT	0%		6%		29
AU17	AAAGGGCCAGGCAT	0%		4%		30
AU18	AATAGGCCCAGACT	0%		7%		31
AU19	AATAGGCCCGCAGT	0%		10%		32
AU20	AAATGGCCGGCAAT	0%		0%		33
GU1	GACGGGCCTAATAT	0%		0%		34
GU2	GCGAGGCCTACTAT	0%		0%		35
GU3	GCAGGGCCTCATTT	0%		0%		36
GU4	GATTGGCCATACAT	0%		0%		37
GU5	GGTTGGCCTAATAT	0%		2%		38
GU6	GGACGGCCAAGCAT	0%		1%		39
GU8	GGCAGGCCTCTTCT	0%		1%		40
GU9	GTTCGGCCCCGACT	0%		0%		41
GU10	GTACGGCCCATAAT	0%		0%		42
GU11	GTATGGCCTCGGAT	0%		0%		43
GU12	GCCGGGCCTTTTTT	0%		7%		44
GU13	GGCCGGCCAAGCAT	0%		6%		45
GU14	GACTGGCCTATAAT	0%		2%		46
GU15	GGACGGCCTACAAT	0%		11%		47

TABLE 3-continued

Name	Sequence	Editing Efficiency				SEQ ID
		no Theo	±SD*	1 mM Theo	±SD*	
GU16	GGGAGGCCAGCGAT	1%		0%		48
GU17	GTAAGGCCGCCGAT	0%		0%		49
GU18	GATTGGCCTACGGT	0%		1%		50
GU19	GATCGGCCATAGAT	3%**	3%	64%**	4%	51
GC1	GCTGGGCCCTTCCC	0%		0%		52
GC2	GTCAGGCCTATACC	0%		0%		53
GC3	GTCCGGCCCCACAC	0%		12%		54
GC4	GGTCGGCCAGTAGC	0%		1%		55
GC5	GTTGGGCCGCGGAC	0%		0%		56
GC6	GATTGGCCAGCAAC	1%		40%		57
GC7	GGGGGGCCGAATAC	1%		71%		58
GC8	GGACGGCCCTGTGC	1%		0%		59
GC9	GCAGGGCCTCTAAC	0%		0%		60
GC10	GAGCGGCCACAACC	0%		0%		61
GC11	GCGTGGCCCTTCCC	0%		3%		62
GC12	GATAGGCCAGTTAC	1%		36%		63
GC13	GGAAGGCCTTATAC	0%		64%		64
GC15	GCATGGCCTACTCC	1%		21%		65
GC16	GATAGGCCAACACC	1%		65%		66
GC17	GACCGGCCCCCGC	0%		2%		67
GC18	GATTGGCCGCAACC	0%		48%		68
GC20	GCAAGGCCAACACC	0%		64%		69

All sequence cassettes for the 2×4 IL and upper stem that were sequenced and tested are listed above with their respective editing efficiency at the galK 1 site when induced for 3 hours with 1 mM theophylline. For the avoidance of doubt, when expressed in a cell, the sequences in Table 3 would be RNA sequences, meaning that each T would be replaced with a U. *±SD represents the standard deviation from the mean. **The mean value was calculated from three biological replicates

TABLE 4

Oligonucleotides	Sequence
theo_insert_B1	GGTATAATACTAGTATGATAAAGCTGCTGCAATAGTTTNNNNNNAT ACCAGCTTCGAAAGAAGCCCTTGGCAG (SEQ ID NO: 70)
theo_Insert_B2	TTTTCAAGTTGATAACGGACTAGCCTTATTTNNNNNNNNCTGCCA AGGGCTTCTTCGAAGCTG (SEQ ID NO: 71)
theo-gRNA_bbF	AAACTATTGCAGCAGCTTTATCATACTAGTATTATACC (SEQ ID NO: 72)
theo-gRNA_bbR	AAAATAAGGCTAGTCCGTTATCAACTTGAAAAAG (SEQ ID NO: 73)
agRNA from Liu et al.	ATGATAAAGCTGCTGCAATAGTTTATAGCTAGAAATAGCAAGTTAA AATAAGGCTAGTCCGTTATCAACTTGAAAAAGTGGACCGAGTCGGT

TABLE 4-continued

	GCCAGCAGCTTTATCATATACCACGCGAAAGCGCCTTGGCAGATGAT AAAGTTTTTTT (SEQ ID NO: 74)
CREATESeq_ fwd	CTAAGGATGATTCTGGAATTC (SEQ ID NO: 75)
CREATESeq_ rev	CAGAACGCAGAAGCG (SEQ ID NO: 76)
guides of the gRNAs	Sequence
galK 1	AUGAUAAGCUGCUGCAAUA (SEQ ID NO: 77)
galK 2	GAUCAGCGGCA AUGUGCCGC (SEQ ID NO: 78)
galK 3	GUUCACCAAUCAAUUCACG (SEQ ID NO: 79)
galK 4	GACCGCGACUCCAGUGAAG (SEQ ID NO: 80)
xylA	CAUAACGAACGCGAUCGAGC (SEQ ID NO: 81)
eGFP (non- targeting)	GACCAGGAUGGGCACCACCC (SEQ ID NO: 82)
mCherry	CCAAGCTGAAGGTGACCAA (SEQ ID NO: 83)
Homology Templates	Sequence
galK 1	CCGCAGGGTGCCGGGTTAAGTTCTTCCGCTTCACTGGAAGTCGCGGT CGGAACGGTATTGCAGCAGCTTTAACATCTGCCGTGGACGCGCAC AAATCGCGCTTAACGGTCAGGAAGCA (SEQ ID NO: 84)
galK 2	GAGCACCGCCTTCGTCTGTCGAGATAGGACATCTGCAACTGCGTAAC AACAGCTTCGGCGCGTGAGCATGGTGATCAGCGGCTAATAACCACA AGGTGCCGGGTTAAGTTCTTCCGCTTCACTGGAAGTCGCGGTCCGAA CCGTA (SEQ ID NO: 85)
galK 3	GAGCACCGCCTTCGTCTGATCTGTAGTCTCTGTTTGCCAACGCATT TGGCTACCTGCCACTCACACATTACAGCGCCTGGCGGTGTTTAAT AAATTGGTGAACACACCGACTACACACGCGTTTCGTTCTGCCCTGC GCGAT (SEQ ID NO: 86)
galK 4	AAGAGCACCGCCTTCGTCTGCTAAGTCTGACTCGGCGCGTGAGCATG GTGATCAGCGGCAATGTGCCGAGGGTGCCGGGTTAAGTTCTTAATA AAGCCTGGAAGTCGCGGTGGAACCGTATTGCAGCAGCTTTATCATC TGCCGC (SEQ ID NO: 87)
xylA	CATCACCGCGGCATTACCTGATTATGGAGTTCAATATGCAAGCCTA TTTTGACCAATTAGACTAATAACGTTATGAAGGCTCAAATCCTCAA ACCCGTTAGCATTCGTCCTACTACA (SEQ ID NO: 88)

DNA and RNA Sequences and oligonucleotides used in this study.

TABLE 5

gRNA construct co-transformed with non-targeting gRNA	# of unedited colonies carrying non-targeting gRNA plasmid
A9 agRNA	9/16
GU19 agRNA	6/16
wt sgRNA	16/16

The over-representation of plasmids expressing non-targeting gRNAs was confirmed by colony PCR and Sanger sequencing. 20 ng CREATE plasmids expressing a non-targeting gRNA were co-transformed with 180 ng of CREATE plasmid expressing a targeting gRNA construct and recovered with 1 mM theophylline. For each of those experiments, 16 colonies that showed up as unedited on the MacConkey agar were picked. For the experiments with A9 and GU19, the different sizes of the non-targeting wt sgRNA and the targeting agRNAs were used for differentiation via

agarose gel electrophoresis after colony PCR. For the experiment with targeting and non-targeting wt sgRNA, Sanger sequencing was used to determine the identity of the transformed plasmid.

Materials and Methods

Strains and Plasmids

For molecular cloning, the *E. coli* strain E. Cloni® was used.

For the recombineering experiments, an *E. coli* MG1655 strain was used that contained two plasmids: The plasmid pSIM5, which expresses the λ -red proteins from a heat shock-inducible promoter and the X2-Cas9 plasmid (Addgene ID: 85811) that encodes Cas9 under control of the arabinose-inducible pBAD promoter. This strain will be referred to as MG λ 9.

The gRNA constructs were constitutively expressed from a CREATE vector that was based on the pUC19 backbone. A CREATE vector also contains a template to introduce mutations at the genomic cut site during homologous repair³⁰.

Molecular Cloning

Homology-based cloning was used for construction of new plasmids. For generating plasmid backbones and inserts

that contained homologous ends, PCR with the Q5® High-Fidelity 2x Master Mix was used. The annealing temperatures typically ranged from 60-72° C. For the cloning of single inserts, CPEC³¹ was used. Libraries were constructed via Gibson Assembly using the NEBuilder® HiFi DNA Assembly Master Mix.

Library Design and Preparation

The double-stranded insert, containing the gRNA constructs, was generated by PCR assembly of two single-stranded oligos. A linearized vector backbone containing homology arms was produced by PCR from the galKOFF plasmid using the primers theo-gRNA_BBF and theo-gRNA_A_BBR. Insert and backbone were assembled by Gibson Assembly. The product was gel purified using a QIAquick Gel Extraction Kit and transformed into *E. Cloni*® bacteria via electroporation. The cells were recovered in SOB media for 1 hour and aliquots were plated on LB agar containing 100 µg/mL carbenicillin, which indicated that about 10⁷ CFU were recovered. The recovery culture was transferred to LB media containing 100 µg/mL carbenicillin, grown for ~14 hours and then 150 µg plasmid library was harvested using a QIAprep Spin Miniprep Kit. Then, 0.5 µg of the plasmid library was electroporated into the MGλ9 strain 24 times to transform 12 µg DNA in total. The bacteria were recovered immediately in LB medium and aliquots were plated on LB agar after 1 hour to estimate the number of CFUs. Then, the MGλ9 culture was subjected to the galK selection assay. The 24 transformations yielded a total MGλ9 CFU of 3×10⁶.

Cloning and Selection of the 14N Library

The first limiting step of the selection was the transformation of the cloning strain with the assembled plasmid library, which yielded about 10⁷ CFUs in total. The transformation of 12 g of this subset of the plasmid library into the MGλ9 strain was expected to yield a total number of CFUs of 6×10⁷, based on control experiments with plasmids that express non-targeting gRNAs. Thus, the sub-library harvested from the *E. Cloni*® is expected to be transformed with 99.8% completion²⁰ into the MGλ9 strain. However, only 3×10⁶ c.f.u.s were actually observed after transformation of MGλ9 with the subset of the plasmid library, which indicates that about 95% of the transformed plasmids probably expressed constitutively active agRNA constructs and the host cells transformed with these constructs were rapidly eliminated from the recovery culture due to cleavage of their genomic DNA.

GalK Selection Assay

Protocols from Warming et al.³³ were mostly followed, with the noticeable difference that the recombineering step was mediated by plasmids, using the CREATE technology, instead of using single-stranded oligos.

Negative selection: The library was transformed into MGλ9 cells without heatshock or addition of gRNA-ligand and recovered at 30° C. for 5 hours in LB containing 0.2% arabinose. The antibiotics chloramphenicol, kanamycin and carbenicillin were added 3 hours after transformation. Constitutively active gRNA constructs that allowed Cas9 activity in the absence of theophylline cause a double-stranded break, which leads to cell death.

Positive selection: The *E. coli* culture underwent recombineering as described in the section "CREATE Recombineering". However, after recovery the culture was not plated on MacConkey, but washed two times in M9 media to remove metabolizable sources of carbon and an aliquot was transferred to M63 selection media, which contained 0.2% glycerol, 0.2% 2-deoxygalactose, 1 mM MgSO₄ and kanamycin and carbenicillin, apart from the M63 salts. This media only permits growth of cells that introduced a stop codon into the galK gene. MG1655 cells typically plateau at a cell density of about OD₆₀₀=2 after 2-3 days in M63

media, so bacteria were added to an initial density of OD₆₀₀<0.2 to allow for enrichment of the editing cells via outgrowth. The cells were grown in the selection media at 37° C. for about 2-3 days or until an OD₆₀₀ of 1.5-2.0 was reached. Then, plasmids of the bacterial culture were harvested and the selection cycle can be repeated for further enrichment. After each transformation step, dilutions of the transformation culture were plated on LB agar to estimate the number of transformants.

Selection conditions were made increasingly stringent with increasing number of cycles by lowering the concentration of gRNA ligand progressively from 1 mM to 250 µM theophylline and by progressively shortening the time available for editing before transfer to selection media from 5 hours to 1 hour. In the third selection cycle, the positive selection was not carried out in liquid culture. Instead, the washed *E. coli* culture was plated on agar plates containing the M63 selection media. After incubating the plates at 37° C. for 2-3 days, colonies were picked from the M63 plates for screening.

CREATE Recombineering

Original Protocol:

The protocol established by Garst et al was mainly followed. LB containing kanamycin and chloramphenicol was inoculated with MGλ9 cells and grown overnight at 30° C. The stationary culture was diluted 100 fold in LB, containing chloramphenicol, kanamycin and 0.2% arabinose and grown to an OD₆₀₀=0.4-0.6 in 25 mL LB. The culture was heat-shocked in a shaking waterbath at 42° C. for 15 minutes to induce expression of the lambda-red proteins from pSIM5. Then, the culture was washed in two volumes of ice-cold deionized water, resuspended in 500 µl deionized water and 50 µl of the cell suspension was electroporated with 200 ng of a CREATE plasmid that encodes gRNA and homology template. The transformants were recovered in 4 mL of LB, containing 0.2% arabinose and optionally 1 mM theophylline or 1 µM anhydrotetracycline (ATC), and shaken at 37° C. for 3 hours before plating on MacConkey Agar. To estimate the number of cfu, aliquots were plated on LB agar 1 hour after transformation.

Aptamer-Protocol:

LB containing kanamycin and chloramphenicol was inoculated with MGλ9 cells and grown overnight at 30° C. The stationary culture was diluted 100 fold in LB, containing chloramphenicol and kanamycin and grown to an OD₆₀₀=0.4-0.6 in 25 mL LB. Then, the culture was washed in two volumes of ice-cold deionized water, resuspended in 500 µl deionized water and 50 µl of the cell suspension was electroporated with 200 ng of a CREATE plasmid that encodes gRNA and homology template. The transformants were recovered in 4 mL of LB, containing 0.2% arabinose and shaken at 30° C. for 1 hour. The culture was heat-shocked in a shaking waterbath at 42° C. for 15 minutes and 1 mM theophylline or 1 µM ATC was added to the culture. The culture was shaken at 37° C. for 3 hours before plating on MacConkey Agar. To estimate the number of cfu, aliquots were plated on LB agar 1 hour after transformation.

Multiplex-Protocol:

For the purpose of first adding theophylline and later 3MX to independently induce two different agRNAs, the protocol was slightly modified. First, the bacteria were prepared and transformed as in the original protocol, 1 mM theophylline was added to the media and the culture was shaken at 30° C. for three hours. Then, the culture was heat-shocked again at 42° C. for 15 minutes to induce expression of the lambda-red proteins again and 1 mM 3MX was added to the media. The culture was shaken at 37° C. for three hours and then plated.

Red/White Screening

In order to assess the frequency of gene editing, a red/white screen was used to visualize the introduction of a stop codon into the *galK* gene by CREATE. The original CREATE protocol was used for inducing gene editing and after 3 hours of recovery (with or without the agRNA ligand) a series of dilutions of the cultures were plated on MacConkey Agar plates containing 1% galactose. Bacterial colonies with a mutated *galK** site appear white on the plates, whereas non-edited colonies appear red. The editing efficiency could be calculated by dividing the number of edited colonies over the total number of colonies. By comparing the editing efficiency in presence versus absence of the ligand, a dynamic range of the in vivo activity could be determined.

When screening the enriched libraries for switchable constructs, the transformants were recovered in 4 mL of LB broth, a dilution series was plated for every construct and the experiment was only carried out once due to the large volume of colonies screened (~250 in total). When the constructs A9 and GU19 were characterized in detail, triplicate of the experiment were carried out. This may account for differences in the observed editing efficiency from the initial screen (FIG. 6) and the values from FIG. 2A.

Calculation of Plasmid Enrichment

When the *E. coli* strain MGλ9 is transformed exclusively with the CREATE *galKOFF* plasmid that expresses wt sgRNA that targets the *galK* gene, 100% of the transformants carry the targeting plasmid and 88.9% of the bacterial population is edited after the CREATE procedure. When transforming a mixture of targeting plasmid and non-targeting plasmid in a 9:1 ratio, 80% of the bacteria are expected to be edited according to $0.9 \times 88.9\% = 80.1\%$, assuming no bias for the non-targeting plasmid. However, only 13.5% of the plated colonies were edited, which means that 15.2% of plated bacteria are expected to carry the targeting CREATE plasmid, when considering that only 88.9% of bacteria that are transformed with the targeting plasmid get edited:

$$\frac{13.5\%}{88.9\%} = 15.2\%.$$

This means, that $100 - 15.2 = 84.8\%$ of the plated bacteria carry the 88.9% non-targeting plasmid, which is a ~8.5-fold enrichment over the 10% of non-targeting plasmid in the transformation mix.

A formalized description would be:

P_{nt} = Fraction of plasmids with non-targeting gRNA

E_t = Editing efficiency of the targeting gRNA

E_{nt} = Editing efficiency obtained from mix with non-targeting plasmids as observed

T_t = Fraction of transformants with the targeting gRNA plasmid

T_{nt} = Fraction of transformants with non-targeting gRNA plasmid

$$\text{Enrichment of non-targeting gRNA plasmid} = \frac{T_{nt}}{P_{nt}} = \frac{1 - T_t}{P_{nt}} = \frac{1 - \left(\frac{E_{nt}}{E_t}\right)}{P_{nt}}$$

The error is given as ± standard deviation.

Cas9 Expression and Purification

Streptococcus pyogenes Cas9 (pMJ915) construct was a gift from Jennifer Doudna (Addgene plasmid #69090)³⁴. The construct was transformed into BL21 (DE3) Rosetta

Escherichia Coli cells. 10 mL LB-Ampicillin bacterial culture was grown overnight and then inoculated into 1 L LB medium. Culture was incubated at 37° C. until OD_{600nm} reached around 0.6. The culture was cooled down to approximately 20° C. in a cold water bath and protein expression was induced by adding 0.5 mM Isopropyl b-D-1-thiogalactopyranoside (IPTG). The culture was grown in a 20° C. shaker overnight. Bacterial cells were pelleted at 1,500 g and resuspended in lysis buffer (1 M KCl, 20 mM HEPES pH 7.5, 20% glycerol, 1 mM TCEP, 10 mM Imidazole). Cells were lysed using an Emulsiflex C3 homogenizer. The cell lysates were clarified by centrifugation at 17,000 g for 30 minutes. Polyethyleneimine (PEI) was used to precipitate the nucleic acid contaminants³⁵. The supernatant (35 mL) was put into a beaker 4° C. and 250 µL 5% PEI was slowly added during stirring. The supernatant was stirred for 15 more minutes. Then, it was centrifuged at 12,000 g for 20 minutes to pellet the nucleic acid contaminants. The supernatant was taken and its PEI concentration was brought to 0.1% and stirred at 4° C. for 15 minutes. It was centrifuged at 12,000 g for 20 minutes. Then, the supernatant was incubated with Ni-NTA sepharose beads on an orbital shaker for 1 hour at 4° C. Beads were centrifuged at 300 g for 2 minutes and washed 3 times in lysis buffer and once in lysis buffer supplemented with 100 mM Imidazole. Proteins were eluted in lysis buffer supplemented with 250 mM Imidazole. The eluate was concentrated and the buffer was exchanged (20 mM HEPES pH 7.5, 500 mM KCl, 1 mM TCEP, 10% glycerol). Then, size exclusion purification was conducted on a Hiload 16/600 Superdex 200 column (AKTA Purifier system (GE Healthcare)). Cas9 protein was purified as monomer, based on comparison to size standards. Final protein concentration was calculated using molar extinction coefficient as determined using the ExPASy-Protparam tool and the absorbance at 280 nm. Cas9 was kept at -20° C.

In Vitro Transcription and Purification of gRNAs

DNA template for RNA transcription was amplified by using PCR and transcribed by T7 RNA polymerase. For a 3 mL transcription reaction, 1.9 mL ddH₂O, 0.3 mL transcription buffer (10×), 100 µL MgCl₂ (1 M), 125 µL from each rNTPs (100 mM), 200 µL PCR template, 31 µL DTT (1 M), 25 µL inorganic pyrophosphatase (20 U/µL), 50 µL T7 RNA polymerase were assembled in a 15 mL canonical. The reaction was vortexed and incubated at 37° C. for 2 hours. Then, 6 mL ethanol was added to the reaction and kept at -80° C. for at least 30 minutes or at 20° C. for overnight to precipitate the RNA. The tubes are centrifuged at 4,000 g and at 4° C. for 15 minutes. The supernatant was discarded and pellet was left for air-drying to evaporate the ethanol at room temperature. The pellet was suspended in 2 mL of 8 M urea, 500 µL 0.5 M EDTA pH 8.0, and 1 mL of formamide loading dye. To re-suspend all of the precipitate the tube was vortexed vigorously. To ensure the complete denaturation of the RNA, samples were heated at 65° C. for 5 minutes and vortexed vigorously until getting a clear solution. Transcripts were purified using denaturing polyacrylamide gel (6-10% 29:1 acrylamide/bisacrylamide, 1×TBE buffer (0.1 M Tris base, 0.8M boric acid, 1 mM Na₂EDTA), 8 M urea). RNA bands were visualized by putting the gel on a fluorescence TLC plate and shadowing the RNA with short-wave UV in a dark room. Full-length transcripts were excised from the gel and the gel pieces were further crushed into small pieces inside a tube. 0.5×TE buffer was added to the tube and the mix was shaken gently at 4° C. for 2 hours to extract the RNA. RNA from the supernatant was concentrated using centrifugal concentrators with a 10 kDa molecu-

lar weight cutoff (Amicon Ultra, 0.5 mL) and buffer (0.5×TE) exchange was performed by the same method. Final RNA concentration was calculated using the absorbance at 260 nm and the molar extinction coefficient as determined using an extinction coefficient calculator that calculates the extinction coefficients by summing of the individual extinction coefficients for each nucleotide in the RNA. The RNA was aliquoted into 5 µl volumes and stored at −20° C. until use.

Body Radiolabeling Reaction of gRNAs

100 µl in vitro RNA transcription reaction was prepared with an adenine ribonucleotide concentration that is 10-fold lower than the standard reaction concentration. 20 µCi ATP [α -32P] was added and the reaction was carried out with T7 RNA polymerase at 37° C. for 2 hours. MicroSpin G25 columns were used to remove unincorporated nucleotides from the labeling reactions. Radiolabeled transcripts were purified using 6% denaturing polyacrylamide gel (29:1 acrylamide/bisacrylamide, 1×TBE buffer, 8 M urea). The gel was exposed using a phosphorimager for about 10-15 minutes and the screen was imaged by using a Typhoon Phosphorimager. The image was printed out with actual sizes. The gel was placed on top of the printed image and the corresponding RNA band was excised from the gel. Gel pieces were put into 2 mL eppendorf tubes and crushed into small pieces by using 1 mL pipette tip. 0.5×TE buffer with 0.3 M

sodium acetate (pH 5.3) was added into the tube and left for elution by rotating at 4° C. for 2 hours. The radiolabeled RNAs were precipitated with ethanol and glycogen at −80° C. for 30 minutes (or overnight at −20° C.) and centrifuged at 17,000 g for 30 minutes at 4° C. Precipitated RNA was resuspended in 0.5×TE buffer and quantified by liquid scintillation counting.

Electrophoretic Mobility Shift Assay (EMSA)

EMSA experiments were carried out to measure dissociation constants (Kd) of the Cas9 binding to gRNAs (FIG. 10) reactions. gRNAs were radiolabeled as described above. They were heated at 95° C. for 3 minutes and snap cooled before addition to the binding reactions. Cas9 proteins (0-5 nM) were incubated with trace amount of (~0.05 nM) radiolabeled RNA molecules in binding buffer containing 20 mM HEPES pH 7.5, 200 mM KCl, 5 mM DTT, 5% glycerol, 0.01% NP40 with or without 250 µM theophylline. A native polyacrylamide (6%, 29:1 acrylamide/bisacrylamide) supplemented with 0.5×TB (45 mM Tris-HCl, 45 mM borate, pH 8.1) buffer was used to separate the bound and unbound gRNA species. Gels were dried and subsequently imaged using a Typhoon Phosphorimager (Molecular Dynamics) and the signals were quantified with ImageQuant software suite. Quantified data was fit to a standard two-state binding isotherm using Igor (Wavemetrics), allowing calculation of both dissociation constants and Hill Coefficients.

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<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 56

gttgggccgc ggac 14

<210> SEQ ID NO 57
<211> LENGTH: 14
<212> TYPE: DNA

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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 57

gattggccag caac 14

<210> SEQ ID NO 58
<211> LENGTH: 14
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 58

ggggggccga atac 14

<210> SEQ ID NO 59
<211> LENGTH: 14
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 59

ggacggccct gtgc 14

<210> SEQ ID NO 60
<211> LENGTH: 14
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 60

gcagggcctc taac 14

<210> SEQ ID NO 61
<211> LENGTH: 14
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 61

gagcgccac aacc 14

<210> SEQ ID NO 62
<211> LENGTH: 14
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 62

gcgtggcct tccc 14

<210> SEQ ID NO 63
<211> LENGTH: 14
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 63

gataggccag ttac 14

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<210> SEQ ID NO 64
<211> LENGTH: 14
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 64

ggaaggcctt atac 14

<210> SEQ ID NO 65
<211> LENGTH: 14
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 65

gcatggccta ctcc 14

<210> SEQ ID NO 66
<211> LENGTH: 14
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 66

gataggccaa cacc 14

<210> SEQ ID NO 67
<211> LENGTH: 14
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 67

gaccggcccc ccgc 14

<210> SEQ ID NO 68
<211> LENGTH: 14
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 68

gattggccgc aacc 14

<210> SEQ ID NO 69
<211> LENGTH: 14
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 69

gcaaggccaa cacc 14

<210> SEQ ID NO 70
<211> LENGTH: 74
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (40)..(45)
<223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 70

ggataataac tagtatgata aagctgctgc aatagttttn nnnnnatacc agcttcgaaa      60
gaagcccttg gcag                                                         74

<210> SEQ ID NO 71
<211> LENGTH: 66
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (34)..(41)
<223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 71

tttttcaagt tgataacgga ctagccttat tttnnnnnnn nctgccaaagg gcttctttcg      60
aagctg                                                         66

<210> SEQ ID NO 72
<211> LENGTH: 39
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 72

aaaactattg cagcagcttt atcatactag tattatacc                            39

<210> SEQ ID NO 73
<211> LENGTH: 34
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 73

aaaataaggc tagtcggtta tcaactgaa aaag                                    34

<210> SEQ ID NO 74
<211> LENGTH: 152
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 74

atgataaagc tgctgcaata gtttttagagc tagaaatagc aagttaaaat aaggctagtc      60
cgttatcaac ttgaaaaagt ggcaccgagt cggtgccagc agctttatca tataccacgc      120
gaaagcgctt tggcagatga taaagttttt tt                                    152

<210> SEQ ID NO 75
<211> LENGTH: 22
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 75

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ctaaggatga tttctggaat tc	22
 <210> SEQ ID NO 76 <211> LENGTH: 15 <212> TYPE: DNA <213> ORGANISM: Artificial Sequence <220> FEATURE: <223> OTHER INFORMATION: Synthetic polynucleotide <400> SEQUENCE: 76	
cagaacgcag aagcg	15
 <210> SEQ ID NO 77 <211> LENGTH: 20 <212> TYPE: RNA <213> ORGANISM: Artificial Sequence <220> FEATURE: <223> OTHER INFORMATION: Synthetic polynucleotide <400> SEQUENCE: 77	
augauaaagc ugcugcaaua	20
 <210> SEQ ID NO 78 <211> LENGTH: 20 <212> TYPE: RNA <213> ORGANISM: Artificial Sequence <220> FEATURE: <223> OTHER INFORMATION: Synthetic polynucleotide <400> SEQUENCE: 78	
gaucagcggc aaugugccgc	20
 <210> SEQ ID NO 79 <211> LENGTH: 20 <212> TYPE: RNA <213> ORGANISM: Artificial Sequence <220> FEATURE: <223> OTHER INFORMATION: Synthetic polynucleotide <400> SEQUENCE: 79	
guucaccaau caaaucacg	20
 <210> SEQ ID NO 80 <211> LENGTH: 20 <212> TYPE: RNA <213> ORGANISM: Artificial Sequence <220> FEATURE: <223> OTHER INFORMATION: Synthetic polynucleotide <400> SEQUENCE: 80	
gaccgcgacu uccagugaag	20
 <210> SEQ ID NO 81 <211> LENGTH: 20 <212> TYPE: RNA <213> ORGANISM: Artificial Sequence <220> FEATURE: <223> OTHER INFORMATION: Synthetic polynucleotide <400> SEQUENCE: 81	
cauaacgaac gcgaucgagc	20
 <210> SEQ ID NO 82 <211> LENGTH: 20 <212> TYPE: RNA	

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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 82

gaccaggaug ggcaccaccc                20

<210> SEQ ID NO 83
<211> LENGTH: 19
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 83

ccaagctgaa ggtgaccaa                19

<210> SEQ ID NO 84
<211> LENGTH: 120
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 84

ccgcaggggtg ccggggttaag ttcttcgct tcactggaag tcgcggtcgg aacggtattg    60
cagcagcttt aacatctgcc gctggacggc gcacaaatcg cgcttaacgg tcaggaagca    120

<210> SEQ ID NO 85
<211> LENGTH: 146
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 85

gagcaccgcc ttcgtctgtc gagataggac atctgcaact gcgtaacaac agcttcggcg    60
gcgtggacat ggtgatcagc ggctaataac cacaaggtgc cgggttaagt tcttcgctt    120
cactggaagt cgcggtcgga accgta                146

<210> SEQ ID NO 86
<211> LENGTH: 146
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 86

gagcaccgcc ttcgtcagat ctgtagtcct ctgtttgcc aacgcatttg ctaccctgcc    60
actcacacca ttcaggcgcc tggccgtgtt taataaattg gtgaacacac cgactacaac    120
gacggtttcg ttctgccctg cgcgat                146

<210> SEQ ID NO 87
<211> LENGTH: 147
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 87

aagagcaccg ccttcgtcgc taagtctgac tcggcggcgt ggacatggtg atcagcgga    60
atgtgccgca gggtgccggg ttaagttctt aataaagcct ggaagtcgcg gtcggaaccg    120

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tattgcagca gctttatcat ctgccgc 147

<210> SEQ ID NO 88
<211> LENGTH: 118
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 88

catcacccgc ggcattacct gattatggag ttcaatatgc aagcctatct tgaccaatta 60

gactaataac gttatgaagg ctcaaatcc tcaaaccgt tagcattccg tcaactaca 118

<210> SEQ ID NO 89
<211> LENGTH: 30
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 89

guuuuagagc uagaaauagc aaguuaaaau 30

<210> SEQ ID NO 90
<211> LENGTH: 45
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (2)..(7)
<223> OTHER INFORMATION: n is a, c, g, or u
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (37)..(44)
<223> OTHER INFORMATION: n is a, c, g, or u

<400> SEQUENCE: 90

unnnnnnaua ccagcuucga aagaagcccu uggcagnnnn nnnna 45

<210> SEQ ID NO 91
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 91

ccgtattgca gcagctttat cat 23

<210> SEQ ID NO 92
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 92

cataacgaac gcgatcgagc tgg 23

<210> SEQ ID NO 93
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 93

cggtattgca gcagctttaa cat                                     23

<210> SEQ ID NO 94
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 94

cataacgtta ttagtctaataa tgg                                   23

<210> SEQ ID NO 95
<211> LENGTH: 29
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 95

auaccagcuu cgaaagaagc ccuuggcag                               29

<210> SEQ ID NO 96
<211> LENGTH: 29
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 96

auaccagcuu cgaaagaagc cauuggcag                               29

<210> SEQ ID NO 97
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 97

guaucgauac cagcuucgaa agaagccnuu ggcagcuuaa gcc             43

<210> SEQ ID NO 98
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 98

agugagauac cagcuucgaa agaagccnuu ggcagcuuaa aa             43

<210> SEQ ID NO 99
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence

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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 99

gagaggauac cagcuucgaa agaagccnuu ggcagccccc ggc          43

<210> SEQ ID NO 100
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 100

ugaaggauac cagcuucgaa agaagccnuu ggcagccgca aca          43

<210> SEQ ID NO 101
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 101

agaaggauac cagcuucgaa agaagccnuu ggcagcccau cau          43

<210> SEQ ID NO 102
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 102

uaguuuauac cagcuucgaa agaagccnuu ggcagaaacc guu          43

<210> SEQ ID NO 103
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 103

ggggggauac cagcuucgaa agaagccnuu ggcagccuau uuu          43

<210> SEQ ID NO 104
<211> LENGTH: 43
<212> TYPE: RNA

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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 104

gauggggauac cagcuucgaa agaagccnuu ggcagccucc acc          43

<210> SEQ ID NO 105
<211> LENGTH: 15
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (7)..(7)
<223> OTHER INFORMATION: n is a, c, g, or u
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 105

ggaggunacg gugcc          15

<210> SEQ ID NO 106
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 106

gagaggauac cagcuucgaa agaagccnuu ggcagccccc ggc          43

<210> SEQ ID NO 107
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 107

cggggggauac cagcuucgaa agaagccnuu ggcagacaau agg          43

<210> SEQ ID NO 108
<211> LENGTH: 15
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (7)..(7)
<223> OTHER INFORMATION: n is a, c, g, or u
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

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<400> SEQUENCE: 108

gggggggncca cgcgc

15

<210> SEQ ID NO 109

<211> LENGTH: 43

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polynucleotide

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (28)..(28)

<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 109

uaguggauac cagcuucgaa agaagccnuu ggcagcuacc aug

43

<210> SEQ ID NO 110

<211> LENGTH: 43

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polynucleotide

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (28)..(28)

<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 110

aggggggauac cagcuucgaa agaagccnuu ggcagccuau aau

43

<210> SEQ ID NO 111

<211> LENGTH: 43

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polynucleotide

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (28)..(28)

<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 111

accaggauac cagcuucgaa agaagccnuu ggcagccaag uau

43

<210> SEQ ID NO 112

<211> LENGTH: 43

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polynucleotide

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (28)..(28)

<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 112

acaaggauac cagcuucgaa agaagccnuu ggcagcccau aau

43

<210> SEQ ID NO 113

<211> LENGTH: 15

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polynucleotide

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (7)..(7)

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<223> OTHER INFORMATION: n is a, c, g, or u
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 113

acucggnccu gaacu 15

<210> SEQ ID NO 114
<211> LENGTH: 15
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (7)..(7)
<223> OTHER INFORMATION: n is a, c, g, or u
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 114

aaggggnccu auaau 15

<210> SEQ ID NO 115
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 115

aguaggauac cagcuucgaa agaagccnuu ggcagccuuu cau 43

<210> SEQ ID NO 116
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 116

acggggauac cagcuucgaa agaagccnuu ggcagccuaa uau 43

<210> SEQ ID NO 117
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 117

aaacggauac cagcuucgaa agaagccnuu ggcagcccac ugu 43

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<210> SEQ ID NO 118
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 118

auagggauac cagcuucgaa agaagccnuu ggcagccauc cau 43

<210> SEQ ID NO 119
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 119

aauaggauac cagcuucgaa agaagccnuu ggcagccacu uau 43

<210> SEQ ID NO 120
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 120

agagggauac cagcuucgaa agaagccnuu ggcagccggg cgu 43

<210> SEQ ID NO 121
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 121

agugggauac cagcuucgaa agaagccnuu ggcagccagc cuu 43

<210> SEQ ID NO 122
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 122

accgggauac cagcuucgaa agaagccnuu ggcagccauu cau 43

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<210> SEQ ID NO 123
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 123

aaccggauac cagcuucgaa agaagccnuu ggcagccccg agu                43

<210> SEQ ID NO 124
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 124

aaagggauac cagcuucgaa agaagccnuu ggcagccagg cau                43

<210> SEQ ID NO 125
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 125

aaauaggauac cagcuucgaa agaagccnuu ggcagcccag acu                43

<210> SEQ ID NO 126
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 126

aaauaggauac cagcuucgaa agaagccnuu ggcagcccg agu                43

<210> SEQ ID NO 127
<211> LENGTH: 15
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (7)..(7)
<223> OTHER INFORMATION: n is a, c, g, or u
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

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<400> SEQUENCE: 127

gguuggnccu aaauau

15

<210> SEQ ID NO 128

<211> LENGTH: 43

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polynucleotide

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (28)..(28)

<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 128

ggacggauac cagcuucgaa agaagccnuu ggcagccaag cau

43

<210> SEQ ID NO 129

<211> LENGTH: 15

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polynucleotide

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (7)..(7)

<223> OTHER INFORMATION: n is a, c, g, or u

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (28)..(28)

<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 129

ggcaggnccu cuucu

15

<210> SEQ ID NO 130

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<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polynucleotide

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (28)..(28)

<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 130

ggcgggauac cagcuucgaa agaagccnuu ggcagccuuu uuu

43

<210> SEQ ID NO 131

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<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polynucleotide

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (28)..(28)

<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 131

ggccggauac cagcuucgaa agaagccnuu ggcagccaag cau

43

<210> SEQ ID NO 132

<211> LENGTH: 15

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (7)..(7)
<223> OTHER INFORMATION: n is a, c, g, or u
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 132

gacuggnccu auaau                                     15

<210> SEQ ID NO 133
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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 133

ggacggauac cagcuucgaa agaagccnuu ggcagccuac aau      43

<210> SEQ ID NO 134
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 134

gauuggauac cagcuucgaa agaagccnuu ggcagccuac ggu      43

<210> SEQ ID NO 135
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 135

gaucggauac cagcuucgaa agaagccnuu ggcagccaua gau      43

<210> SEQ ID NO 136
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 136

guccggauac cagcuucgaa agaagccnuu ggcagcccca cac      43

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<210> SEQ ID NO 137
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
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<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 137

ggucggauac cagcuucgaa agaagccnuu ggcagccagu agc 43

<210> SEQ ID NO 138
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 138

gauuggauac cagcuucgaa agaagccnuu ggcagccagc aac 43

<210> SEQ ID NO 139
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 139

ggggggauac cagcuucgaa agaagccnuu ggcagccgaa uac 43

<210> SEQ ID NO 140
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 140

gcguggauac cagcuucgaa agaagccnuu ggcagcccuu ccc 43

<210> SEQ ID NO 141
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 141

gauaggauac cagcuucgaa agaagccnuu ggcagccagu uac 43

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<210> SEQ ID NO 142
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 142

ggaaggauac cagcuucgaa agaagccnuu ggcagccuua uac

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43

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<210> SEQ ID NO 143
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 143

gcauggauac cagcuucgaa agaagccnuu ggcagccuac ucc

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43

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<210> SEQ ID NO 144
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 144

gauaggauac cagcuucgaa agaagccnuu ggcagccaac acc

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43

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<210> SEQ ID NO 145
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 145

gaccggauac cagcuucgaa agaagccnuu ggcagccccc cgc

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43

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<210> SEQ ID NO 146
<211> LENGTH: 15
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (7)..(7)
<223> OTHER INFORMATION: n is a, c, g, or u
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

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<400> SEQUENCE: 146

gauuggnccg caacc

15

<210> SEQ ID NO 147

<211> LENGTH: 43

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polynucleotide

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (28)..(28)

<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 147

gcaaggauac cagcuucgaa agaagccnuu ggcagccaac acc

43

<210> SEQ ID NO 148

<211> LENGTH: 43

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polynucleotide

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (28)..(28)

<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 148

guaucgauac cagcuucgaa agaagccnuu ggcagcuuua gcc

43

<210> SEQ ID NO 149

<211> LENGTH: 43

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polynucleotide

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (28)..(28)

<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 149

gagaggauac cagcuucgaa agaagccnuu ggcagccccc ggc

43

<210> SEQ ID NO 150

<211> LENGTH: 43

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polynucleotide

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (28)..(28)

<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 150

ugaaggauac cagcuucgaa agaagccnuu ggcagccgca aca

43

<210> SEQ ID NO 151

<211> LENGTH: 43

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polynucleotide

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (28)..(28)

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<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 151

uaguuuauac cagcuucgaa agaagccnuu ggcagaaacc guu

43

<210> SEQ ID NO 152

<211> LENGTH: 42

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polynucleotide

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (27)..(27)

<223> OTHER INFORMATION: n is a, c, g, or u

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (28)..(28)

<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 152

gggggauacc agcuucgaaa gaagccnuug gcagccuauu uu

42

<210> SEQ ID NO 153

<211> LENGTH: 43

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polynucleotide

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (28)..(28)

<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 153

ggaggauuac cagcuucgaa agaagccnuu ggcagacggu gcc

43

<210> SEQ ID NO 154

<211> LENGTH: 43

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polynucleotide

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (28)..(28)

<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 154

gagaggauac cagcuucgaa agaagccnuu ggcagccccc ggc

43

<210> SEQ ID NO 155

<211> LENGTH: 43

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic polynucleotide

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (28)..(28)

<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 155

cgggggauac cagcuucgaa agaagccnuu ggcagacaau agg

43

<210> SEQ ID NO 156

<211> LENGTH: 43

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 156

gggggggauac cagcuucgaa agaagccnuu ggcagccacg cgc 43

<210> SEQ ID NO 157
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (7)..(7)
<223> OTHER INFORMATION: n is the small-molecule-binding aptamer
sequence
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 157

uaguggauac cagcuucgaa agaagccnuu ggcagcuacc aug 43

<210> SEQ ID NO 158
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 158

acaaggauac cagcuucgaa agaagccnuu ggcagcccau aau 43

<210> SEQ ID NO 159
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 159

acucggauac cagcuucgaa agaagccnuu ggcagccuga acu 43

<210> SEQ ID NO 160
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 160

aaggggauac cagcuucgaa agaagccnuu ggcagccuau aau 43

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<210> SEQ ID NO 161
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 161

aguaggauac cagcuucgaa agaagccnuu ggcagccuuu cau 43

<210> SEQ ID NO 162
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 162

acggggauac cagcuucgaa agaagccnuu ggcagccuaa uau 43

<210> SEQ ID NO 163
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 163

gaucggauac cagcuucgaa agaagccnuu ggcagccaua gau 43

<210> SEQ ID NO 164
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 164

ggggggauac cagcuucgaa agaagccnuu ggcagccgaa uac 43

<210> SEQ ID NO 165
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 165

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ggaaggauac cagcuucgaa agaagccnuu ggcagccuua uac 43

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<210> SEQ ID NO 166
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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 166

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gauaggauac cagcuucgaa agaagccnuu ggcagccaac acc 43

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<210> SEQ ID NO 167
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 167

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gcaaggauac cagcuucgaa agaagccnuu ggcagccaac acc 43

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<210> SEQ ID NO 168
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (2)..(4)
<223> OTHER INFORMATION: n is A, C, G, or U
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (38)..(42)
<223> OTHER INFORMATION: n is A, C, G or U

<400> SEQUENCE: 168

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gnnncgauac cagcuucgaa agaagccnuu ggcagcunnn nnc 43

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<210> SEQ ID NO 169
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polynucleotide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (2)..(4)
<223> OTHER INFORMATION: n is A, C, G, or U
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (38)..(42)
<223> OTHER INFORMATION: n is A, C, G, or U

<400> SEQUENCE: 169

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 gnnnggauac cagcuucgaa agaagccnuu ggcagccnnn nnc

43

<210> SEQ ID NO 170
 <211> LENGTH: 43
 <212> TYPE: RNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polynucleotide
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (2)..(4)
 <223> OTHER INFORMATION: n is A, C, G, or U
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (28)..(28)
 <223> OTHER INFORMATION: n is A or C
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (38)..(42)
 <223> OTHER INFORMATION: n is A, C, G, or U

 <400> SEQUENCE: 170

gnnnggauac cagcuucgaa agaagccnuu ggcagccnnn nnu

43

<210> SEQ ID NO 171
 <211> LENGTH: 43
 <212> TYPE: RNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polynucleotide
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (2)..(4)
 <223> OTHER INFORMATION: n is A, C, G, or U
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (28)..(28)
 <223> OTHER INFORMATION: n is A or C
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (37)..(42)
 <223> OTHER INFORMATION: n is A, C, G, or U

 <400> SEQUENCE: 171

gnnnggauac cagcuucgaa agaagccnuu ggcagannnn nnc

43

<210> SEQ ID NO 172
 <211> LENGTH: 43
 <212> TYPE: RNA
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 <223> OTHER INFORMATION: n is A or C
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (38)..(42)
 <223> OTHER INFORMATION: n is A, C, G, or U

 <400> SEQUENCE: 172

annnggauac cagcuucgaa agaagccnuu ggcagccnnn nnu

43

<210> SEQ ID NO 173
 <211> LENGTH: 43
 <212> TYPE: RNA

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<223> OTHER INFORMATION: n is A or C
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<222> LOCATION: (37)..(42)
<223> OTHER INFORMATION: n is A, C, G, or U

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<400> SEQUENCE: 173

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cnnnggauac cagcuucgaa agaagccnuu ggcagannnn nng

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43

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<210> SEQ ID NO 174
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<212> TYPE: RNA
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<222> LOCATION: (2)..(4)
<223> OTHER INFORMATION: n is A, C, G, or U
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(28)
<223> OTHER INFORMATION: n is A or C
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (36)..(40)
<223> OTHER INFORMATION: n is A, C, G, or U

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<400> SEQUENCE: 174

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unnnggauac cagcuucgaa agaagccnuu ggcagnnnnn a

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41

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<210> SEQ ID NO 175
<211> LENGTH: 41
<212> TYPE: RNA
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<220> FEATURE:
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<222> LOCATION: (2)..(4)
<223> OTHER INFORMATION: n is A, C, G, or U
<220> FEATURE:
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<220> FEATURE:
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<400> SEQUENCE: 175

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unnnggauac cagcuucgaa agaagccnuu ggcagnnnnn g

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41

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<210> SEQ ID NO 176
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<223> OTHER INFORMATION: n is A, C, G, or U

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<223> OTHER INFORMATION: n is A, C, G, or U

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<400> SEQUENCE: 176

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unnnuuauac cagcuucgaa agaagccnuu ggcagnnnnn u

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41

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<210> SEQ ID NO 177
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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
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<223> OTHER INFORMATION: n is A, C, G, or U
<220> FEATURE:
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<223> OTHER INFORMATION: n is A or C
<220> FEATURE:
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<222> LOCATION: (38)..(42)
<223> OTHER INFORMATION: n is A, C, G, or U

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<400> SEQUENCE: 177

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gnnnggauac cagcuucgaa agaagccnuu ggcagccnnn nnu

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43

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<210> SEQ ID NO 178
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<212> TYPE: RNA
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<222> LOCATION: (1)..(6)
<223> OTHER INFORMATION: n is A, C, G, or U
<220> FEATURE:
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<223> OTHER INFORMATION: n is A or C
<220> FEATURE:
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<223> OTHER INFORMATION: n is A, C, G, or U
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<223> OTHER INFORMATION: n is a, c, g, or u

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<400> SEQUENCE: 178

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nnnnnnauac cagcuucgaa agaagccnuu ggcagnnnnn nnn

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43

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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Cas9 amino acid sequence

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<400> SEQUENCE: 179

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Met Asp Lys Lys Tyr Ser Ile Gly Leu Asp Ile Gly Thr Asn Ser Val
1           5           10           15

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```

Gly Trp Ala Val Ile Thr Asp Glu Tyr Lys Val Pro Ser Lys Lys Phe

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-continued

20					25					30					
Lys	Val	Leu	Gly	Asn	Thr	Asp	Arg	His	Ser	Ile	Lys	Lys	Asn	Leu	Ile
		35					40					45			
Gly	Ala	Leu	Leu	Phe	Asp	Ser	Gly	Glu	Thr	Ala	Glu	Ala	Thr	Arg	Leu
		50				55					60				
Lys	Arg	Thr	Ala	Arg	Arg	Arg	Tyr	Thr	Arg	Arg	Lys	Asn	Arg	Ile	Cys
65					70					75					80
Tyr	Leu	Gln	Glu	Ile	Phe	Ser	Asn	Glu	Met	Ala	Lys	Val	Asp	Asp	Ser
				85					90					95	
Phe	Phe	His	Arg	Leu	Glu	Glu	Ser	Phe	Leu	Val	Glu	Glu	Asp	Lys	Lys
			100					105					110		
His	Glu	Arg	His	Pro	Ile	Phe	Gly	Asn	Ile	Val	Asp	Glu	Val	Ala	Tyr
		115					120					125			
His	Glu	Lys	Tyr	Pro	Thr	Ile	Tyr	His	Leu	Arg	Lys	Lys	Leu	Val	Asp
		130				135					140				
Ser	Thr	Asp	Lys	Ala	Asp	Leu	Arg	Leu	Ile	Tyr	Leu	Ala	Leu	Ala	His
145					150					155					160
Met	Ile	Lys	Phe	Arg	Gly	His	Phe	Leu	Ile	Glu	Gly	Asp	Leu	Asn	Pro
			165					170						175	
Asp	Asn	Ser	Asp	Val	Asp	Lys	Leu	Phe	Ile	Gln	Leu	Val	Gln	Thr	Tyr
			180					185					190		
Asn	Gln	Leu	Phe	Glu	Glu	Asn	Pro	Ile	Asn	Ala	Ser	Gly	Val	Asp	Ala
		195				200						205			
Lys	Ala	Ile	Leu	Ser	Ala	Arg	Leu	Ser	Lys	Ser	Arg	Arg	Leu	Glu	Asn
	210					215					220				
Leu	Ile	Ala	Gln	Leu	Pro	Gly	Glu	Lys	Lys	Asn	Gly	Leu	Phe	Gly	Asn
225					230					235					240
Leu	Ile	Ala	Leu	Ser	Leu	Gly	Leu	Thr	Pro	Asn	Phe	Lys	Ser	Asn	Phe
			245					250						255	
Asp	Leu	Ala	Glu	Asp	Ala	Lys	Leu	Gln	Leu	Ser	Lys	Asp	Thr	Tyr	Asp
		260					265						270		
Asp	Asp	Leu	Asp	Asn	Leu	Leu	Ala	Gln	Ile	Gly	Asp	Gln	Tyr	Ala	Asp
		275				280						285			
Leu	Phe	Leu	Ala	Ala	Lys	Asn	Leu	Ser	Asp	Ala	Ile	Leu	Leu	Ser	Asp
	290					295					300				
Ile	Leu	Arg	Val	Asn	Thr	Glu	Ile	Thr	Lys	Ala	Pro	Leu	Ser	Ala	Ser
305				310						315					320
Met	Ile	Lys	Arg	Tyr	Asp	Glu	His	His	Gln	Asp	Leu	Thr	Leu	Leu	Lys
			325						330					335	
Ala	Leu	Val	Arg	Gln	Gln	Leu	Pro	Glu	Lys	Tyr	Lys	Glu	Ile	Phe	Phe
			340					345					350		
Asp	Gln	Ser	Lys	Asn	Gly	Tyr	Ala	Gly	Tyr	Ile	Asp	Gly	Gly	Ala	Ser
		355					360					365			
Gln	Glu	Glu	Phe	Tyr	Lys	Phe	Ile	Lys	Pro	Ile	Leu	Glu	Lys	Met	Asp
	370					375					380				
Gly	Thr	Glu	Glu	Leu	Leu	Val	Lys	Leu	Asn	Arg	Glu	Asp	Leu	Leu	Arg
385					390					395					400
Lys	Gln	Arg	Thr	Phe	Asp	Asn	Gly	Ser	Ile	Pro	His	Gln	Ile	His	Leu
			405						410					415	
Gly	Glu	Leu	His	Ala	Ile	Leu	Arg	Arg	Gln	Glu	Asp	Phe	Tyr	Pro	Phe
			420					425					430		
Leu	Lys	Asp	Asn	Arg	Glu	Lys	Ile	Glu	Lys	Ile	Leu	Thr	Phe	Arg	Ile
	435						440					445			

-continued

Pro	Tyr	Tyr	Val	Gly	Pro	Leu	Ala	Arg	Gly	Asn	Ser	Arg	Phe	Ala	Trp
450						455					460				
Met	Thr	Arg	Lys	Ser	Glu	Glu	Thr	Ile	Thr	Pro	Trp	Asn	Phe	Glu	Glu
465					470					475				480	
Val	Val	Asp	Lys	Gly	Ala	Ser	Ala	Gln	Ser	Phe	Ile	Glu	Arg	Met	Thr
				485					490					495	
Asn	Phe	Asp	Lys	Asn	Leu	Pro	Asn	Glu	Lys	Val	Leu	Pro	Lys	His	Ser
		500						505					510		
Leu	Leu	Tyr	Glu	Tyr	Phe	Thr	Val	Tyr	Asn	Glu	Leu	Thr	Lys	Val	Lys
		515				520						525			
Tyr	Val	Thr	Glu	Gly	Met	Arg	Lys	Pro	Ala	Phe	Leu	Ser	Gly	Glu	Gln
	530					535					540				
Lys	Lys	Ala	Ile	Val	Asp	Leu	Leu	Phe	Lys	Thr	Asn	Arg	Lys	Val	Thr
545					550					555				560	
Val	Lys	Gln	Leu	Lys	Glu	Asp	Tyr	Phe	Lys	Lys	Ile	Glu	Cys	Phe	Asp
			565						570					575	
Ser	Val	Glu	Ile	Ser	Gly	Val	Glu	Asp	Arg	Phe	Asn	Ala	Ser	Leu	Gly
		580						585					590		
Thr	Tyr	His	Asp	Leu	Leu	Lys	Ile	Ile	Lys	Asp	Lys	Asp	Phe	Leu	Asp
		595				600						605			
Asn	Glu	Glu	Asn	Glu	Asp	Ile	Leu	Glu	Asp	Ile	Val	Leu	Thr	Leu	Thr
	610					615					620				
Leu	Phe	Glu	Asp	Arg	Glu	Met	Ile	Glu	Glu	Arg	Leu	Lys	Thr	Tyr	Ala
625					630					635					640
His	Leu	Phe	Asp	Asp	Lys	Val	Met	Lys	Gln	Leu	Lys	Arg	Arg	Arg	Tyr
			645						650					655	
Thr	Gly	Trp	Gly	Arg	Leu	Ser	Arg	Lys	Leu	Ile	Asn	Gly	Ile	Arg	Asp
		660						665					670		
Lys	Gln	Ser	Gly	Lys	Thr	Ile	Leu	Asp	Phe	Leu	Lys	Ser	Asp	Gly	Phe
	675						680					685			
Ala	Asn	Arg	Asn	Phe	Met	Gln	Leu	Ile	His	Asp	Asp	Ser	Leu	Thr	Phe
	690					695					700				
Lys	Glu	Asp	Ile	Gln	Lys	Ala	Gln	Val	Ser	Gly	Gln	Gly	Asp	Ser	Leu
705					710					715					720
His	Glu	His	Ile	Ala	Asn	Leu	Ala	Gly	Ser	Pro	Ala	Ile	Lys	Lys	Gly
			725					730						735	
Ile	Leu	Gln	Thr	Val	Lys	Val	Val	Asp	Glu	Leu	Val	Lys	Val	Met	Gly
		740						745					750		
Arg	His	Lys	Pro	Glu	Asn	Ile	Val	Ile	Glu	Met	Ala	Arg	Glu	Asn	Gln
		755				760						765			
Thr	Thr	Gln	Lys	Gly	Gln	Lys	Asn	Ser	Arg	Glu	Arg	Met	Lys	Arg	Ile
	770				775						780				
Glu	Glu	Gly	Ile	Lys	Glu	Leu	Gly	Ser	Gln	Ile	Leu	Lys	Glu	His	Pro
785					790					795					800
Val	Glu	Asn	Thr	Gln	Leu	Gln	Asn	Glu	Lys	Leu	Tyr	Leu	Tyr	Tyr	Leu
				805					810					815	
Gln	Asn	Gly	Arg	Asp	Met	Tyr	Val	Asp	Gln	Glu	Leu	Asp	Ile	Asn	Arg
			820					825					830		
Leu	Ser	Asp	Tyr	Asp	Val	Asp	His	Ile	Val	Pro	Gln	Ser	Phe	Leu	Lys
	835						840					845			
Asp	Asp	Ser	Ile	Asp	Asn	Lys	Val	Leu	Thr	Arg	Ser	Asp	Lys	Asn	Arg
	850					855					860				

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Gly	Lys	Ser	Asp	Asn	Val	Pro	Ser	Glu	Glu	Val	Val	Lys	Lys	Met	Lys	865	870	875	880
Asn	Tyr	Trp	Arg	Gln	Leu	Leu	Asn	Ala	Lys	Leu	Ile	Thr	Gln	Arg	Lys	885	890	895	
Phe	Asp	Asn	Leu	Thr	Lys	Ala	Glu	Arg	Gly	Gly	Leu	Ser	Glu	Leu	Asp	900	905	910	
Lys	Ala	Gly	Phe	Ile	Lys	Arg	Gln	Leu	Val	Glu	Thr	Arg	Gln	Ile	Thr	915	920	925	
Lys	His	Val	Ala	Gln	Ile	Leu	Asp	Ser	Arg	Met	Asn	Thr	Lys	Tyr	Asp	930	935	940	
Glu	Asn	Asp	Lys	Leu	Ile	Arg	Glu	Val	Lys	Val	Ile	Thr	Leu	Lys	Ser	945	950	955	960
Lys	Leu	Val	Ser	Asp	Phe	Arg	Lys	Asp	Phe	Gln	Phe	Tyr	Lys	Val	Arg	965	970	975	
Glu	Ile	Asn	Asn	Tyr	His	His	Ala	His	Asp	Ala	Tyr	Leu	Asn	Ala	Val	980	985	990	
Val	Gly	Thr	Ala	Leu	Ile	Lys	Lys	Tyr	Pro	Lys	Leu	Glu	Ser	Glu	Phe	995	1000	1005	
Val	Tyr	Gly	Asp	Tyr	Lys	Val	Tyr	Asp	Val	Arg	Lys	Met	Ile	Ala		1010	1015	1020	
Lys	Ser	Glu	Gln	Glu	Ile	Gly	Lys	Ala	Thr	Ala	Lys	Tyr	Phe	Phe		1025	1030	1035	
Tyr	Ser	Asn	Ile	Met	Asn	Phe	Phe	Lys	Thr	Glu	Ile	Thr	Leu	Ala		1040	1045	1050	
Asn	Gly	Glu	Ile	Arg	Lys	Arg	Pro	Leu	Ile	Glu	Thr	Asn	Gly	Glu		1055	1060	1065	
Thr	Gly	Glu	Ile	Val	Trp	Asp	Lys	Gly	Arg	Asp	Phe	Ala	Thr	Val		1070	1075	1080	
Arg	Lys	Val	Leu	Ser	Met	Pro	Gln	Val	Asn	Ile	Val	Lys	Lys	Thr		1085	1090	1095	
Glu	Val	Gln	Thr	Gly	Gly	Phe	Ser	Lys	Glu	Ser	Ile	Leu	Pro	Lys		1100	1105	1110	
Arg	Asn	Ser	Asp	Lys	Leu	Ile	Ala	Arg	Lys	Lys	Asp	Trp	Asp	Pro		1115	1120	1125	
Lys	Lys	Tyr	Gly	Gly	Phe	Asp	Ser	Pro	Thr	Val	Ala	Tyr	Ser	Val		1130	1135	1140	
Leu	Val	Val	Ala	Lys	Val	Glu	Lys	Gly	Lys	Ser	Lys	Lys	Leu	Lys		1145	1150	1155	
Ser	Val	Lys	Glu	Leu	Leu	Gly	Ile	Thr	Ile	Met	Glu	Arg	Ser	Ser		1160	1165	1170	
Phe	Glu	Lys	Asn	Pro	Ile	Asp	Phe	Leu	Glu	Ala	Lys	Gly	Tyr	Lys		1175	1180	1185	
Glu	Val	Lys	Lys	Asp	Leu	Ile	Ile	Lys	Leu	Pro	Lys	Tyr	Ser	Leu		1190	1195	1200	
Phe	Glu	Leu	Glu	Asn	Gly	Arg	Lys	Arg	Met	Leu	Ala	Ser	Ala	Gly		1205	1210	1215	
Glu	Leu	Gln	Lys	Gly	Asn	Glu	Leu	Ala	Leu	Pro	Ser	Lys	Tyr	Val		1220	1225	1230	
Asn	Phe	Leu	Tyr	Leu	Ala	Ser	His	Tyr	Glu	Lys	Leu	Lys	Gly	Ser		1235	1240	1245	
Pro	Glu	Asp	Asn	Glu	Gln	Lys	Gln	Leu	Phe	Val	Glu	Gln	His	Lys		1250	1255	1260	
His	Tyr	Leu	Asp	Glu	Ile	Ile	Glu	Gln	Ile	Ser	Glu	Phe	Ser	Lys					

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1265	1270	1275
Arg Val Ile Leu Ala Asp	Ala Asn Leu Asp Lys	Val Leu Ser Ala
1280	1285	1290
Tyr Asn Lys His Arg Asp	Lys Pro Ile Arg Glu Gln	Ala Glu Asn
1295	1300	1305
Ile Ile His Leu Phe Thr	Leu Thr Asn Leu Gly Ala	Pro Ala Ala
1310	1315	1320
Phe Lys Tyr Phe Asp Thr	Thr Ile Asp Arg Lys Arg	Tyr Thr Ser
1325	1330	1335
Thr Lys Glu Val Leu Asp	Ala Thr Leu Ile His Gln	Ser Ile Thr
1340	1345	1350
Gly Leu Tyr Glu Thr Arg	Ile Asp Leu Ser Gln Leu	Gly Gly Asp
1355	1360	1365

<210> SEQ ID NO 180
 <211> LENGTH: 47
 <212> TYPE: RNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polynucleotide

<400> SEQUENCE: 180

gagaggugaa gaauacgacc accuagguag aaauaccuaa aacauac

47

<210> SEQ ID NO 181
 <211> LENGTH: 29
 <212> TYPE: RNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic - Aptamer 1/2 Consensus
 <220> FEATURE:
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 <222> LOCATION: (22)..(22)
 <223> OTHER INFORMATION: n is A or C

<400> SEQUENCE: 181

auaccagcuu cgaaagaagc cnuuggcag

29

<210> SEQ ID NO 182
 <211> LENGTH: 43
 <212> TYPE: RNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
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 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (2)..(4)
 <223> OTHER INFORMATION: n is a, c, g, or u
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (38)..(42)
 <223> OTHER INFORMATION: n is a, c, g, or u

<400> SEQUENCE: 182

gnnnggauac cagcuucgaa agaagccuu ggcagccnnn nnc

43

<210> SEQ ID NO 183
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 <212> TYPE: RNA
 <213> ORGANISM: Artificial Sequence
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<220> FEATURE:
<221> NAME/KEY: misc_feature
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<400> SEQUENCE: 183

annnggauac cagcuucgaa agaagcccuu ggcagccnnn nnu 43

<210> SEQ ID NO 184
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<212> TYPE: RNA
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<223> OTHER INFORMATION: Synthetic polynucleotide
<220> FEATURE:
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<222> LOCATION: (2)..(4)
<223> OTHER INFORMATION: n is a, c, g, or u
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<222> LOCATION: (38)..(42)
<223> OTHER INFORMATION: n is a, c, g, or u

<400> SEQUENCE: 184

gnnnggauac cagcuucgaa agaagcccuu ggcagccnnn nnu 43

<210> SEQ ID NO 185
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<212> TYPE: RNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 185

agagcuauac cagcuucgaa agaagcccuu ggcagagcaa guu 43

<210> SEQ ID NO 186
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 186

ugaaggauac cagcuucgaa agaagcccuu ggcagccgca aca 43

<210> SEQ ID NO 187
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 187

gaucggauac cagcuucgaa agaagcccuu ggcagccaaa gau 43

<210> SEQ ID NO 188
<211> LENGTH: 43
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic polynucleotide

<400> SEQUENCE: 188

ggggggauac cagcuucgaa agaagcccuu ggcagccuau uuu 43

What is claimed is:

1. A guide RNA (sgRNA) comprising a small-molecule-binding aptamer sequence adjacent to a region comprising a 2×4 internal loop and a region comprising an upper stem, wherein the small-molecule-binding aptamer is a theophylline-binding aptamer and the theophylline-binding aptamer comprises at least 85% homology to a polynucleotide represented by the sequence 5'-AUACCAGCUUCGA AAGAAGCCCUUGGCAG-3' (SEQ ID NO: 95).

2. The sgRNA of claim 1, wherein sgRNA comprises up to 2 nucleotides between the small-molecule-binding aptamer sequence and the 2×4 internal loop.

3. A ribonucleoprotein (RNP) comprising an sgRNA of claim 1 and a Cas enzyme.

4. The RNP of claim 3, wherein the Cas enzyme is Cas9.

5. The RNP of claim 3, wherein the region comprising the 2×4 internal loop and the region comprising the upper stem comprises the sequence of SEQ ID NO: 169.

6. The RNP of claim 3, wherein the region comprising the 2×4 internal loop and the region comprising the upper stem comprises the sequence of SEQ ID NO: 100.

7. A method of inducing gene editing in a bacterium, the method comprising:

- a. introducing or expressing the sgRNA of claim 1 in a bacterium,
- b. introducing or expressing a Cas enzyme in the bacterium, and
- c. contacting the bacterium with a small molecule that interacts with the small-molecule-binding aptamer sequence in the sgRNA such that gene editing is induced in the bacterium.

8. The method of claim 7, wherein the Cas enzyme is Cas9.

9. The method of claim 7, wherein the sgRNA exhibits greater than or equal to 10-fold induction in editing in presence of a small molecule ligand that binds to the small-molecule-binding aptamer sequence.

10. An in vivo method of selecting in bacteria a switchable aptamer single guide RNA (agRNA) that is inducible with a small molecule, wherein the switchable agRNA comprises a single guide RNA of claim 1, the method comprising:

- a. performing a negative selection step comprising growing a first plurality of bacteria in absence of the small molecule and in the absence of X-red protein expression, inducing cell death in a subset of the plurality of bacteria to produce a second plurality of bacteria, wherein individual bacteria in the first plurality of bacteria comprise a nucleic acid encoding the candidate agRNA comprising the sgRNA of claim 1 that produces a cut site in a gene encoding a selection marker protein: a Cas enzyme; and a template for homologous repair of the cut site in the gene encoding the selection marker protein;
- b. performing a positive selection step comprising either:
 - i. growing the second plurality of bacteria in the presence of the small molecule and red protein expression; or
 - ii. isolating nucleic acids encoding the candidate agRNAs comprising the sgRNA of claim 1 from the second plurality of bacteria and introducing the nucleic acids into a third plurality of bacteria and growing the third plurality of bacteria in the presence of the small molecule and X-red protein expression, wherein individual bacteria in the third plurality comprise a nucleic acid encoding a candidate agRNA that produces a cut site in the gene encoding a selection marker protein, a Cas

enzyme, and a template for homologous repair of the cut site in the gene that encodes the selection marker protein; and

- c. selecting at least one bacterium from b. that expresses an altered selection marker protein.

11. The method of claim 10, wherein the selection marker protein is galK and the template for homologous repair comprises a premature stop codon for a galK gene.

12. An in vivo method of selecting in bacteria a switchable aptamer single guide RNA (agRNA) that is inducible with a small molecule, wherein the switchable agRNA comprises a single guide RNA of claim 1, the method comprising:

- a. performing a negative selection step comprising growing a first plurality of bacteria in absence of the small molecule and in presence of X-red protein expression and inducing cell death in a subset of a first plurality of bacteria to produce a second plurality of bacteria, wherein individual bacteria in the first plurality of bacteria comprise a nucleic acid encoding the candidate agRNA comprising the sgRNA of claim 1 that produces a cut site in a gene encoding a selection marker protein, a Cas enzyme, and a template for homologous repair of the cut site in the gene that encodes the selection marker protein;
- b. performing a positive selection step comprising either:
 - i. growing the second plurality of bacteria in the presence of the small molecule and X-red protein expression; or
 - ii. isolating nucleic acids encoding candidate agRNAs from the second plurality of bacteria and introducing the nucleic acids into a third plurality of bacteria and growing the third plurality of bacteria in the presence of the small molecule and λ-red protein expression, wherein individual bacteria in the third plurality comprise a nucleic acid encoding the candidate agRNAs comprising the sgRNA of claim 1 that produces a cut site in the gene encoding a selection marker protein, a Cas enzyme, and a template for homologous repair of the cut site in the gene that encodes the selection marker protein; and
- c. selecting at least one bacterium from b. that expresses the selection marker protein.

13. The sgRNA of claim 1, wherein the region comprising the 2×4 internal loop and the upper stem comprises the sequence of SEQ ID NO: 169.

14. A composition comprising: (i) the sgRNA of claim 1; or (ii) a vector expressing the sgRNA of claim 1.

15. The sgRNA of claim 1, wherein the small-molecule-binding aptamer is a theophylline-binding aptamer, and the theophylline-binding aptamer comprises the polynucleotide represented by the sequence 5'-AUACCAGCUUCG AAAGAAGCCCUUGGCAG-3' (SEQ ID NO: 95).

16. A guide RNA (sgRNA) comprising a small-molecule-binding aptamer sequence adjacent to a region comprising a 2×4 internal loop and a region comprising an upper stem, wherein the region comprising the 2×4 internal loop and the upper stem comprises the sequence 5'-GNNNGGAUAC CAGCUUCGAAAGAAGCCNUUGGCAGCCNNNNNC-3' (SEQ ID NO: 169).

17. The sgRNA of claim 16, wherein sgRNA comprises up to 2 nucleotides between the small-molecule-binding aptamer sequence and the 2×4 internal loop.

18. A guide RNA (sgRNA) comprising a small-molecule-binding aptamer sequence adjacent to a region comprising a 2×4 internal loop and a region comprising an upper stem, wherein the region comprising the 2×4 internal loop and the upper stem comprises a polynucleotide represented by SEQ

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ID NO: 100, UGAAGGAUACCAGCUUCGAAAG
AAGCCNUUGGCAGCCGCAACA.

19. The sgRNA of claim **18**, wherein sgRNA comprises
up to 2 nucleotides between the small-molecule-binding
aptamer sequence and the 2×4 internal loop. 5

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